



SINGAPORE
MANAGEMENT
UNIVERSITY

ISSS602: Data Analytics Lab

Assignment 2: Hospitality Segmentation with Cluster Analysis

12 September 2025

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Estimated word count for content (not counting captions, references, content page, etc): ~3600

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1. Objective

The objective of this report is to segment Northern Adriatic coastal accommodations into defensible clusters using rigorously prepared data and SAS Viya analyses (EDA, variable reduction, and multiple clustering methods), and translate the results into concise, evidence-based profiles and recommendations.

2. Data Pre-Checks

2.1. Basic Checks

```
19  /* 1) Schema inventory */
20  proc contents data=&LIB..&CHA out=work._CHA_COLS noprint; run;
21
22  proc sql noprint;
23      select count(*) into :_n_rows trimmed from &LIB..&CHA;
24      select count(*) into :_n_cols   trimmed from work._CHA_COLS;
25      select count(*) into :_nnum     trimmed from work._CHA_COLS where type=1;
26      select count(*) into :_nchar    trimmed from work._CHA_COLS where type=2;
27
28      /* Build var lists to loop over safely */
29      select name into :_numlist separated by ' '
30      from work._CHA_COLS where type=1 order by varnum;
31
32      select name into :_charlist separated by ' '
33      from work._CHA_COLS where type=2 order by varnum;
34  quit;
35
36  %put NOTE: &LIB..&CHA rows=&n_rows cols=&n_cols (num=&nnum char=&nchar).;
37
```

Figure 1: Code snippet of basic checks (Assign2/1_DataChecks/1_DataPreChecks.sas)

Top 30 variables by missing rate

varname	type	n_miss	n_total	miss_rate
DIST_ESTUARY	N	1	5030	0.02%
AREA	N	0	5030	0.00%
DIST_SEA	N	0	5030	0.00%
DIST_WATERWAY	N	0	5030	0.00%
ENCODED_ACCESS	N	0	5030	0.00%
ENCODED_ECOCERTIFD	N	0	5030	0.00%
ENCODED_GREEN	N	0	5030	0.00%
ENCODED_POOL	N	0	5030	0.00%
ENCODED_SUSTAINABL	N	0	5030	0.00%
ENCODED_TYPE	N	0	5030	0.00%
LATITUDE	N	0	5030	0.00%
LONGITUDE	N	0	5030	0.00%

Figure 2. Result of Basic Checks

The initial audit isn't accurate because it only counted true SAS missings (numeric . and blank character values). It did not treat common placeholder tokens ("NA", "N/A", "NULL", ".", "-", "UNKNOWN") as missing, nor numeric sentinels (e.g., -1) that mean

“unknown.” As a result, the “only one missing in DIST_ESTUARY and 0% elsewhere” finding understates real missingness and gives a false sense of completeness.

2.2. Hygiene & Normalization

```

37  /* 2a) Numeric diagnostics (levels, missing, integerness via MOD, -1) */
38  macro build_num_diag;
39      %if &_nnum>0 %then %do;
40          %local i var sep;
41          proc sql;
42              create table work._NUM_DIAG as
43              %let sep=;
44              %do i=1 %to &_nnum;
45                  %let var=%scan(&_numlist,&i);
46                  %sep
47                  select upcase("&var") as varname length=32
48                      , 'N' as stype length=1
49                      , count(*) as n_total_n
50                      , sum(missing(&var)) as n_miss_n
51                      , (calculated n_total_n - calculated n_miss_n) as n_nonmiss_n
52                      , count(distinct &var) as nlevels_n
53                      , sum( case when not missing(&var) and mod(&var,1) ne 0 then 1 else 0 end ) as n_nonint
54                      , sum(&var = -1) as n_neg1
55                  from &LIB..&CHA
56                  %let sep=union all;
57              %end;
58          ;
59          quit;
60          proc sort data=work._NUM_DIAG; by varname; run;
61      %end;
62  %mend;
63  %build_num_diag
64  ..

```

Figure 3. Code snippet for Hygiene & Normalization(Assign2/1_DataChecks/2_HygieneNormalization)

Hygiene dashboard

check	value
Variables (total)	19
Numeric vars	12
Character vars	7
Vars w/ -1 present (numeric)	5
Vars w/ placeholder tokens (char)	6
Recommend = Category	9
Recommend = Measure	6
Recommend = Label	4

Numeric with -1 present (potential 'unknown' codes)

varname	n_neg1	n_total	neg1_rate
ENCODED_SUSTAINABL	1214	5030	24.14%
ENCODED_GREEN	361	5030	7.18%
ENCODED_ACCESS	360	5030	7.16%
ENCODED_POOL	358	5030	7.12%
ENCODED_ECOCERTIFD	318	5030	6.32%

Character with placeholder tokens (NA/#N/A/NULL/etc.)

varname	n_placeholders	n_total	placeholder_rate
RATINGS	1395	5030	27.73%
ROOMS	1372	5030	27.28%
PRICE	875	5030	17.40%
SLR_MEDIAN_RCP45_2030	834	5030	16.58%
HEIGHT	742	5030	14.75%
GROND_Z	470	5030	9.34%

Figure 4. Results for Hygiene and Normalization Checks

The six geospatial/area distances behave as clean continuous measures. Practically, the encoded_* variables should be treated as categorical (with an “Unknown” level for -1), normalize placeholders to missing, parse character-numerics to real numbers, and then proceed with necessary steps for it to be data-ready for clustering analysis.

2.3. Keys & Geo Basics

```

124  /* 2) Exact duplicates (by full-row hash) */
125  ⊖ proc sql;
126      create table _EXACT_GRP as
127      select _row_hash, count(*) as n_rows
128      from _C3_BASE
129      group by _row_hash
130      having calculated n_rows>1
131      order by n_rows desc;
132
133      create table _DUP_EXACT as
134      select b.*
135      from _C3_BASE b
136      inner join _EXACT_GRP g
137      on b._row_hash = g._row_hash
138      order by g.n_rows desc;
139  quit;
140
141  /* 3) Near duplicates (rounded lat/lon clusters) */
142  ⊖ proc sql;
143      create table _NEAR_GRP as
144      select _cluster, count(*) as n_rows
145      from _C3_BASE
146      where not missing(_cluster)
147      group by _cluster
148      having calculated n_rows>1
149      order by n_rows desc;
150
151      /* Note: select b.* only (avoid duplicate _cluster warning) */
152      create table _NEAR_DUPS as
153      select b.*
154      from _C3_BASE b
155      inner join _NEAR_GRP g
156      on b._cluster = g._cluster
157      order by g.n_rows desc, b._cluster;
158  quit;

```

Figure 5. Code snippet of keys & geo basics. (Assign2/1_DataChecks/3_KeysGeoBasics.sas)

KEY_GEO_SUMMARY

metric	value
Rows (total)	5030
Candidate key unique	4116
Candidate key duplicated	457
Exact duplicate groups	0
Exact duplicate rows	0
Near-dup clusters (4d.p.)	690
Near-dup rows	1512
BAD lat/lon rows	0

Figure 6. Result from the Key & Geo Checks

The key and geospatial checks indicate generally clean coordinates but meaningful duplication pressure. Out of 5,030 records, the candidate key (NAME + latitude + longitude) is unique for 4,116 keys, while 457 keys recur; this implies about 457 “excess” rows (~9.1%) beyond a one-row-per-establishment baseline. There are 690 near-duplicate clusters covering 1,512 rows (~30.1%; ~2.19 rows per cluster), consistent with geocoding jitter or multi-listing of the same place.

2.4. Metadata Level Conformance

```

67  /* Join counts to labels via the map (varname -> Column) */
68  ⊖ proc sql;
69      /* Detailed table */
70      create table LEVEL_CONFORMANCE as
71      select k.varname,
72             c.code_num as code,
73             coalesce(m.label, cats('UNMAPPED code=',strip(put(c.code_num,best.)))) as label length=200,
74             c.n
75      from _obs_counts c
76      inner join map_keep k
77           on upcase(c.varname)=upcase(k.varname)
78      left join md_codes m
79           on upcase(k.col)=m.col and c.code_num=m.code_num
80      order by k.varname, label;
81
82      /* Simple counts by label */
83      create table LEVEL_COUNTS as
84      select varname, label, sum(n) as n
85      from LEVEL_CONFORMANCE
86      group by varname, label
87      order by varname, n desc, label;
88  quit;

```

Figure 7. Code snippet of Metadata's Conformance Checks(Assign2/1_DataChecks/4_MetadataConformance.sas)

varname	label	n
encoded_Access	No	2201
encoded_Access	Yes	1453
encoded_Access	Yes (Private)	1016
encoded_Access	NA	360
encoded_EcoCertifd	FALSE	4595
encoded_EcoCertifd	NA	318
encoded_EcoCertifd	TRUE	117
encoded_Green	No	3957
encoded_Green	Yes	712
encoded_Green	NA	361
encoded_Pool	No	3400
encoded_Pool	Yes	1250
encoded_Pool	NA	358
encoded_Pool	Yes (Private)	22
encoded_Type	hotel	3479
encoded_Type	holiday_rental_home	1286
encoded_Type	bed_and_breakfast	93
encoded_Type	camping	76
encoded_Type	guesthouse	73
encoded_Type	agriturismo	14
encoded_Type	hostel	8
encoded_Type	UNMAPPED code=8	1

```

95  /* Checking unmapped Type */
96  ⊖ proc sql;
97      create table type_stray as
98      select *
99      from casuser.cha_raw
100     where encoded_Type not in (
101         select Encoded from casuser.metadata_cha
102         where upcase(Column)='TYPE'
103     );
104  quit;

```

Figure 8. Results of the metadata conformance checks and code to check for unmapped type.

From the checks, there is one single unmapped code in encoded_Type where there is an 8 not listed in the metadata. Upon running further code, this belongs to the hotel named 'Hotel Arizona'. This will be taken into consideration when performing analysis.

2.5. Variables Quick Look (Summary)

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/* numeric targets */
length
grond_z_n slr_median_rcp45_2030_n dist_sea_n dist_waterway_n dist_estuary_n
height_n area_n Rooms_n Price_n Ratings_n ;

/* helper for char+num with NA/blank → missing */
length _t $64;
macro to_num(src, dst);
  if vtype($src)='C' then do;
    _t = strip($src);
    if missing(_t) or upcase(_t)='NA' then &dst = .;
    else &dst = input(_t, ?? best32.);
  end;
else &dst = $src;
%mend;

%to_num(grond_z,      grond_z_n);
%to_num(slr_median_rcp45_2030, slr_median_rcp45_2030_n);
%to_num(dist_sea,     dist_sea_n);
%to_num(dist_waterway, dist_waterway_n);
%to_num(dist_estuary, dist_estuary_n);
%to_num(height,       height_n);
%to_num(area,         area_n);
%to_num(Rooms,        Rooms_n);
%to_num(Price,        Price_n);
%to_num(Ratings,      Ratings_n);

/* replace ONLY in the NEW table */
drop grond_z slr_median_rcp45_2030 dist_sea dist_waterway dist_estuary
height area Rooms Price Ratings _t;
rename
  grond_z_n      = grond_z
  slr_median_rcp45_2030_n = slr_median_rcp45_2030
  dist_sea_n      = dist_sea
  dist_waterway_n = dist_waterway
  dist_estuary_n  = dist_estuary
  height_n        = height
  area_n          = area
  Rooms_n         = Rooms
  Price_n         = Price
  Ratings_n       = Ratings
;
run;

```

Figure 9. Code snippet to change categorical variables to numeric measures.
(Assign2/1_DataChecks/5_VisualizationPreChecks)

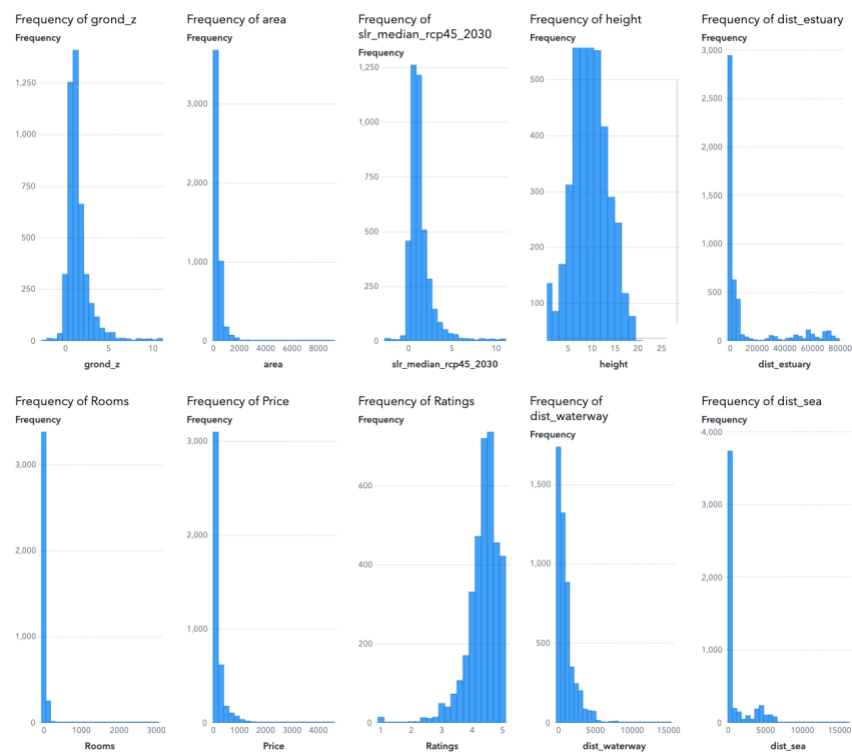


Figure 10. Histogram of variables (Assign2/Reports/1_QuickLook)

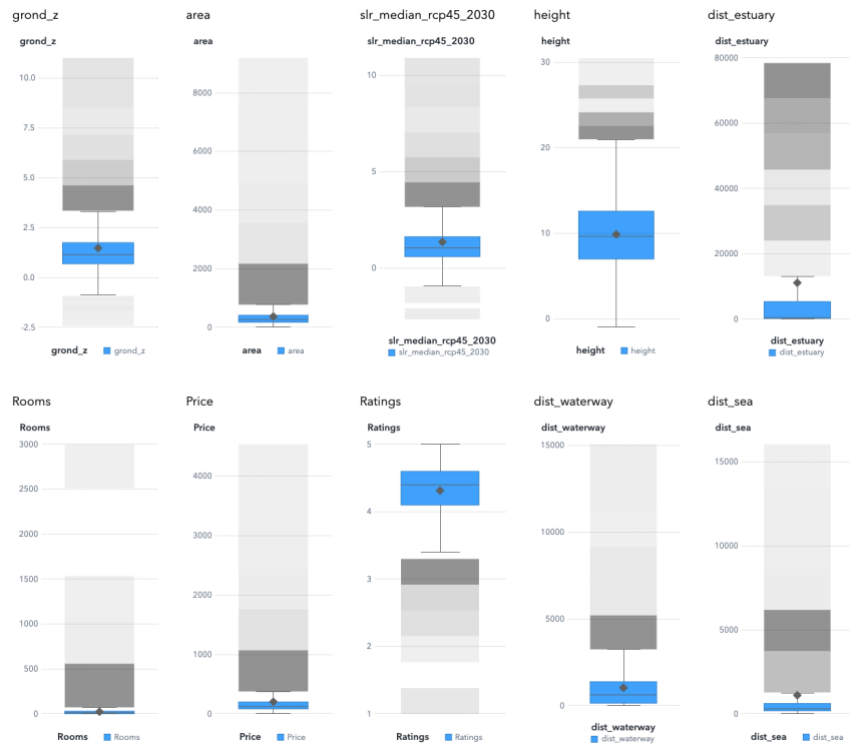


Figure 11. Box Plots of Variables(Assign2/Reports/1_QuickLook)

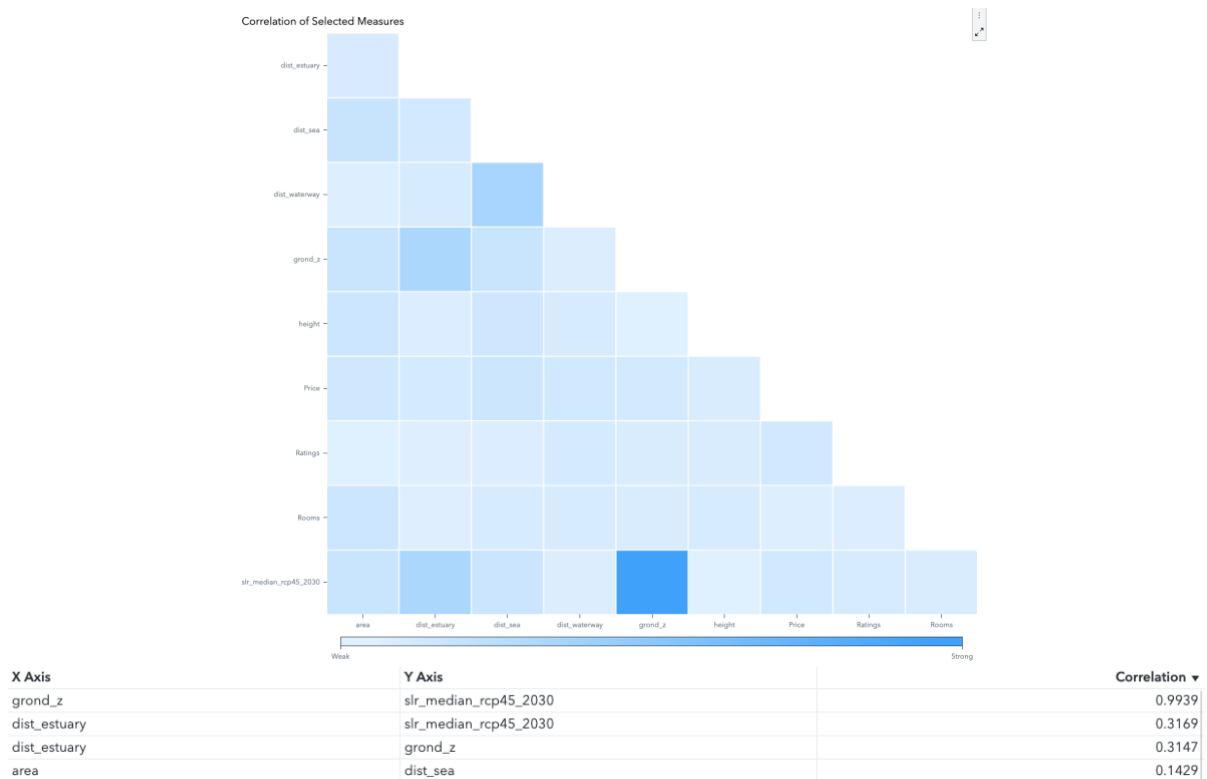
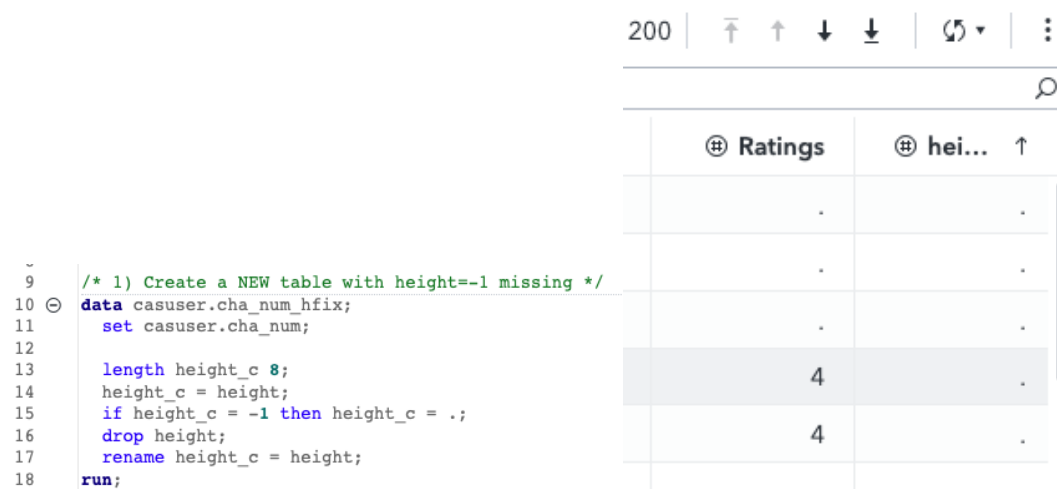


Figure 12. Correlation Matrix between Variables, results arrange in descending correlation (Assign2/Reports/1_QuickLook)

Distance measures are strongly right-skewed with many high-end outliers, indicating the need for standardisation and, if redundancy arises, a single coastal-proximity variable. The two elevation variables (grond_z and slr_median_rcp45_2030) are near-perfectly collinear (≈ 0.994), so only one should enter clustering to avoid double-counting. Physical-scale indicators show heavy right tails (Rooms, area) and mostly low values with a thin upper tail for height, again supporting standardisation and optional log-type sensitivity checks for Rooms/area. Market/quality variables diverge: Price is right-skewed; Ratings are bounded with a ceiling near 5, so both can be included but standardised. Apart from the elevation pair, correlations remain below the usual $|r|=0.85$ threshold, supporting a compact 6–7 variable set.

3. Data Preparation

3.1. Defusing Dummy Data: Height



```

9      /* 1) Create a NEW table with height=-1 missing */
10  data casuser.cha_num_hfix;
11  set casuser.cha_num;
12
13  length height_c 8;
14  height_c = height;
15  if height_c = -1 then height_c = .;
16  drop height;
17  rename height_c = height;
18  run;

```

	⊕ Ratings	⊕ hei... ↑
	.	.
	.	.
	4	.
	4	.

Figure 13. Code snippet of replacing -1 in height and its results(Assign2/2_DataPreparation/1_HeightFix)

From the pre-checks done, it was discovered that in the source data, there are 5 rows with height value of -1 which is a sentinel indicating “unknown,” not a physically meaningful measurement. Leaving -1 as numeric would create artificial low outliers, distort summary statistics and z-scores, and risk spurious separation in distance-based clustering. Replacing height = -1 with missing (NA) preserves the measurement scale, enables appropriate imputation alongside genuine missings, and prevents the placeholder from influencing cluster geometry.

3.2. Exposure Gap (Collinearity between 2 Variables)

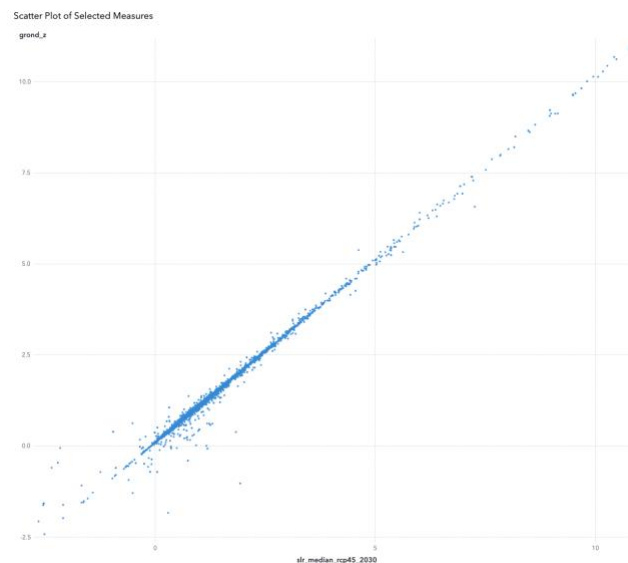


Figure 14. Scatter plot of *grond_z* vs *slr_median_rcp45_2030*

The scatter hugs the 45° line, confirming the two measures (*grond_z* and *slr_median_rcp45_2030*) are near-identical across properties.

```

1  /* Exposure Gap */
2  data casuser.cha_num_tf;
3  set casuser.cha_num_tf;
4
5  length exposure_gap_m $ gap_le_0 3 gap_bin $;
6
7  /* Vertical safety margin (m) */
8  if missing(grond_z) or missing(slr_median_rcp45_2030) then exposure_gap_m = .;
9  else exposure_gap_m = grond_z - slr_median_rcp45_2030;
10
11 /* Red flag: at/under projected 2030 level */
12 gap_le_0 = (not missing(exposure_gap_m) and exposure_gap_m <= 0);
13
14 /* Coarse bins for profiling (NOT used in clustering distance) */
15 if missing(exposure_gap_m) then gap_bin = .;
16 else if exposure_gap_m <= 0 then gap_bin = 0; /* At/under */
17 else if exposure_gap_m < 1 then gap_bin = 1; /* 0<1 m */
18 else if exposure_gap_m < 3 then gap_bin = 2; /* 1<3 m */
19 else gap_bin = 3; /* ≥3 m */
20
21 run;
22
23 /* Labels */
24 proc format;
25 value gap_bin;
26 0 = '≤ 0 m (at/under)';
27 1 = '0<1 m';
28 2 = '1<3 m';
29 3 = '≥ 3 m';
30 run;
31
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63
64 /* 1) High-resolution look at the gap distribution */
65 proc univariate data=casuser.cha_num_tf;
66 var exposure_gap_m;
67 histogram exposure_gap_m / endpoints=-2 to 2 by 0.05 kernel;
68 inset n nmiss mean std pl p5 p25 median p75 p95 min max / position=ne;
69 qqplot exposure_gap_m / normal(mu=est sigma=est);
70 title "Exposure gap (m): fine bins + kernel + quantiles";
71 run;
72
73
74 /* 2) How close is grond_z to slr_2030? (slope, intercept, R^2, VIF) */
75 proc reg data=casuser.cha_num_tf;
76 model grond_z = slr_median_rcp45_2030 / vif;
77 title "grond_z ~ slr_2030 (slope/intercept/R^2/VIF)";
78 run; quit;
79
80
81 /* 3) Summaries */
82 proc means data=casuser.cha_num_tf;
83 n nmiss mean std min pl p5 p25 median p75 p95 max;
84 var grond_z slr_median_rcp45_2030 exposure_gap_m;
85 title "Elevation, 2030 SLR (as given), and exposure gap: spread check";
86 run;
87
88
89
90
91
92
93
94
95
96
97
98
99
100

```

Figure 15. Code snippet of *grond_z* and *slr_median_rcp45_2030* and its checks. (Assign2/2_DataPreparation/2_ExposureGap.sas)

Moments			
N	4196	Sum Weights	4196
Mean	0.09120859	Sum Observations	382.711255
Std Deviation	0.15534931	Variance	0.02413341
Skewness	-5.9304573	Kurtosis	147.629939
Uncorrected SS	136.146205	Corrected SS	101.23965
Coeff Variation	170.323111	Std Error Mean	0.00239823

Basic Statistical Measures		
Location		Variability
Mean	0.091209	Std Deviation 0.15535
Median	0.097997	Variance 0.02413
Mode	0.073985	Range 5.06443
		Interquartile Range 0.02623

Figure 16. Result of checks

Regression shows $\text{grond_z} \approx \text{slr_2030}$ (slope 0.995, intercept 0.10, R^2 0.988). Consequently, the exposure gap ($\text{grond_z} - \text{slr_2030}$) is near zero for most properties (SD 0.16 m, IQR 0.026 m, ~16.6% missing) with only a few extremes, so as a

standardized clustering driver, it would add little discrimination and risk amplifying noise. The decision is to use `grond_z` as the sole vertical driver; exclude `slr_median_rcp45_2030`.

3.3. Log Transformation

```

1  /* 1) Create transformed copy with log features (originals preserved) */
2  ⊖ data casuser.cha_num_tf;
3    set casuser.cha_num_hfix;
4    rooms_log = log(1 + Rooms);
5    area_log  = log(1 + area);
6    price_log = log(1 + Price);
7  run;

```

Figure 17. Code snippet of log transformation for Rooms, Area, and Price
(Assign2/2_DataPreparation/3_LogTransformation.sas)

Distribution Summary: Originals vs Log-Transformed

The MEANS Procedure

Variable	N	N Miss	Minimum	25th Pctl	Median	75th Pctl	Maximum	Skewness
Rooms	3658	1372	1.0000000	2.0000000	17.0000000	30.0000000	3000.00	27.7280305
area	5030	0	6.8170769	175.9315949	271.4113919	416.1805474	9191.60	7.4501017
Price	4155	875	1.0000000	84.0000000	122.0000000	201.0000000	4536.00	5.7036799
rooms_log	3658	1372	0.6931472	1.0986123	2.8903718	3.4339872	8.0067008	-0.0564757
area_log	5030	0	2.0563107	5.1757632	5.6073134	6.0335191	9.1261540	-0.1072144
price_log	4155	875	0.6931472	4.4426513	4.8121844	5.3082677	8.4200213	0.7274117

Distribution Summary: Other Clustering Measures

The MEANS Procedure

Variable	N	N Miss	Minimum	25th Pctl	Median	75th Pctl	Maximum
dist_sea	5030	0	5.9709196	186.9358318	300.8624397	622.4804358	16041.22
slr_median_rcp45_2030	4196	834	-2.6453624	0.5995918	1.0567126	1.6308208	10.9017390
height	4283	747	0	7.0000000	9.6518000	12.5917000	30.5136000
Ratings	3635	1395	1.0000000	4.1000000	4.4000000	4.6000000	5.0000000

Figure 18. Result of log transformation

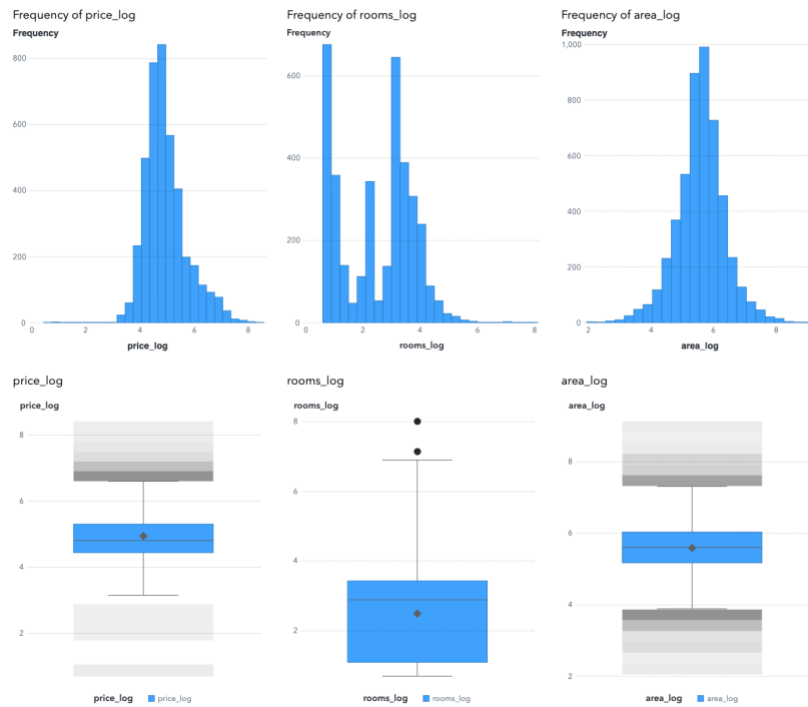


Figure 19. Histogram and box plot of log-transformed variables

Rooms, area, and Price were log transformed to tame their extreme right-skew and outliers so Euclidean distances aren't dominated by a few very large properties/prices; the transform makes equal relative changes comparable. On the data it works as intended: skewness falls from 27.73 to -0.06 (Rooms), 7.45 to -0.11 (area), and 5.70 to 0.73 (Price), collapsing long right tails into near-symmetric shapes. The log transformed variables will feed the log features into clustering after standardizing them so each has comparable weight, and original units are kept for clear, human-readable profiling.

3.4. Z-Scoring

Variable	N	N Miss	Mean	Std Dev
dist_sea_z	5030	0	3.365543E-16	1.0000000
dist_waterway_z	5030	0	3.930212E-15	1.0000000
dist_estuary_z	5029	1	5.746933E-16	1.0000000
grond_z_z	4560	470	4.140937E-16	1.0000000
area_log_z	5030	0	-3.47544E-14	1.0000000
price_log_z	4155	875	-2.48865E-14	1.0000000
height_z	4283	747	-1.20691E-15	1.0000000
rooms_log_z	3658	1372	-2.79041E-14	1.0000000

```

18 /* 2) Z-score ONLY the *_z variables.
19 Missing inputs remain missing; non-missing are standardized. */
20 proc stdize data=casuser.cha_full_zprep
21 out=casuser.cha_full_std
22 method=std;
23 var dist_sea_z dist_waterway_z dist_estuary_z
24 grond_z_z area_log_z price_log_z height_z rooms_log_z;
25 run;
26

```

Figure 20. Code snippet and results of Z-scoring variables(Assign2/2_DataPreparation/4_ZScoring.sas)

The clustering work relies that is about to be done relies on Euclidean distance, which is highly scale-sensitive. Z-scoring puts every driver on a common scale(mean 0, SD 1), so they contribute comparably instead of letting the largest- variance variable dominate. This complements earlier log transforms. Medians on the distance z-scores are slightly negative, consistent with long right tails (a few properties very far from water pull the mean upward). Z-scores were created for all potential drivers so the data is cluster-ready, but only a selected subset will enter the distance metric.

3.5. Final Missingness Check

17	/* 1) Numeric variables: treat SAS-missing OR (encoded_* = -1) as missing *	variable	N total	N missing	% missing
18	%if %sysevalf(&N_NUM > 0) %then %do;	Ratings	5030	1395	27.73
19	data work_miss_num(keep=variable n_missing);	Rooms	5030	1372	27.28
20	set casuser.cha_full_std end=last;	rooms_log	5030	1372	27.28
21	array nvars _numeric_;	rooms_log_z	5030	1372	27.28
22	array cnts[&N_NUM] _temporary_;	encoded_Sustainabl	5030	1214	24.14
23	length _nm \$32;	Price	5030	875	17.40
24		price_log	5030	875	17.40
25	do _i=1 to dim(nvars);	price_log_z	5030	875	17.40
26	_nm = upcase(vname(nvars[_i]));	exposure_gap_m	5030	834	16.58
27	if missing(nvars[_i])	gap_bin	5030	834	16.58
28	or (substr(_nm,1,8)='ENCODED_' and nvars[_i]=-1)	slr_median_rcp45_2030	5030	834	16.58
29	then cnts[_i] + 1;	height	5030	747	14.85
30	end;	height_z	5030	747	14.85
31		grond_z	5030	470	9.34
32	if last then do _i=1 to dim(nvars);	grond_z_z	5030	470	9.34
33	variable = vname(nvars[_i]);	encoded_Green	5030	361	7.18
34	n_missing = cnts[_i];	encoded_Access	5030	360	7.16
35	output;	encoded_Pool	5030	358	7.12
36	end;	encoded_EcoCertifd	5030	318	6.32
37	run;	dist_estuary	5030	1	0.02
38	%end;	dist_estuary_z	5030	1	0.02
39	%else %do;				
40	data work_miss_num; length variable \$32; n_missing=.; stop; run;				
41	%end;				

Figure 21. Code snippet of final missingness check and its results (Assign2/2_DataPreparation/5_MissingCheck.sas)

The first pre-check only counted true SAS missings (numeric . / blank char) and therefore understated gaps where placeholders exist (notably encoded_* = -1 and earlier NA-like tokens). A complete-case policy is adopted for clustering drivers: when a driver has < 20% missing, drivers with NA will be dropped. In the data this applies to price_log (17.4%), height (14.85%), and grond_z (9.34%), while dist_sea, dist_waterway, and area_log have 0% missing. This choice avoids injecting artificial structure into variables that drive Euclidean distance, and keeps the math transparent.

3.6. Duplicate rows

Duplicate Summary (counts and % of total)

Metric	Count	Percent
Total rows	5,030	.
Rows missing any key part	0	0.00%
Rows in duplicate key groups	914	18.17%
Duplicate rows beyond first (key)	457	9.09%
Exact duplicate rows (beyond first)	0	0.00%

```

32 proc sql;
33 create table work_dup_keys as
34 select &KEY1 as key1 format=$200.,
35        &KEY2 as key2,
36        &KEY3 as key3,
37        count(*) as n
38 from work.&TAB
39 where not missing(&KEY1) and nmiss(&KEY2, &KEY3)=0
40 group by &KEY1, &KEY2, &KEY3
41 having calculated n > 1
42 order by n desc, &KEY1, &KEY2, &KEY3;
43 quit;

```

Figure 22. Duplicate check results and code snippets (Assign2/2_DataPreparation/6B_DuplicateChecks.sas)

Obs	Name	latitude	longitude	encoded_Type	encoded_Access	encoded_EcoCertif	encoded_Green	encoded_Pool	encoded_Sustainabl	grond_z	slr_median_rcp45_2030	dist_sea	dist_waterway	dist_estuary	area	Rooms	Price	Rating
1	a tribute to music	45.4339	12.3470	7	1	1	1	1	1	0.83765	0.73800	3418.66	99.06	3418.66	166.59	12	121	4.3
2	a tribute to music	45.4339	12.3470	7	1	1	1	1	1	0.83765	0.73800	3418.66	99.06	3418.66	496.78	12	121	4.3
3	affittacamere giabetta	44.2528	12.3594	5	2	1	2	1	-1	0.96642	0.86114	510.47	1395.67	510.47	124.01	.	.	.
4	affittacamere giabetta	44.2528	12.3594	5	2	1	2	1	-1	0.99950	0.90894	510.47	1395.67	510.47	161.44	.	.	.
5	ai mori d'oriente	45.4460	12.3304	7	1	1	1	1	1	0.66758	0.67142	5363.00	24.77	5363.00	393.17	51	341	4.7
6	ai mori d'oriente	45.4460	12.3304	7	1	1	1	1	1	0.66758	0.67142	5363.00	24.77	5363.00	198.46	51	341	4.7
7	ai patrizi venezia	45.4353	12.3346	5	1	1	1	1	1	0.98510	0.89329	4425.36	70.16	4425.36	1495.54	11	351	4.8
8	ai patrizi venezia	45.4353	12.3346	5	1	1	1	1	1	0.98555	0.89291	4425.36	70.16	4425.36	535.58	11	351	4.8
9	al redentore di venezia	45.4256	12.3292	7	1	1	1	1	1	0.79024	0.67385	4075.89	120.20	4075.89	350.25	8	225	4.7
10	al redentore di venezia	45.4256	12.3292	7	1	1	1	1	1	0.79024	0.67385	4075.89	120.20	4075.89	102.26	8	225	4.7

Figure 23. Sample duplicate rows where all the values are the same.

Row counts (BEFORE, AFTER, REMOVED)

BEFORE	AFTER	REMOVED
5030	4573	457

Figure 24. Duplicates removed sanity check (Assign2/2_DataPreparation/6B_RemovingDuplicates)

The dataset was audited again, this time for duplicate listings using the key Name + latitude + longitude and found 457 redundant rows (9.09% of 5,030). Each duplicate group was collapsed to a single record by ranking rows on completeness of core drivers/profilers. Post-checks confirmed that all unique keys were preserved and only extras were removed (5,030 → 4,573; removed = 457). This prevents double-weighting the same property and keeps the geographic/elevation/market signal clean for clustering.

3.7. Clustering Variables

*_z denotes variables that were z-scored during data preparation.

1. **dist_sea_z** (distance to sea, z-scored): primary coastal attractiveness and surge-exposure proxy.
2. **dist_waterway_z** (distance to river/canal, z-scored): fluvial/canal pathways and waterfront appeal distinct from the open sea. It is only modestly correlated with dist_sea_z, so it contributes new geometric signal.
3. **grond_z_z** (ground elevation, z-scored): vertical safety. Prefer elevation in the distance metric and reserve derived exposure constructs (e.g., exposure_gap_m, gap_le_0) for post-hoc risk storytelling to avoid redundancy.
4. **area_log_z** (log area, z-scored): asset scale and CAPEX proxy; log transformation stabilizes heavy right tails.
5. **price_log_z** (log price, z-scored): market position/pricing power; log stabilizes skew and preserves rank order.
6. **height_z** (building height, z-scored): structural verticality relevant to both risk mitigation and urban form.

Why not include dist_estuary in the distance?

Estuary proximity partly overlaps the elevation/proximity geometry and is less uniformly interpretable at network scale. Retaining it for profiling (not distance) keeps the clustering space parsimonious and avoids over-weighting coastal geometry.

Why not include rooms_log, Ratings, and encoded_*?

Their higher missingness and/or high dimensionality risk diluting the core geometry. It will be deliberately reserved for post-cluster validation and naming, which also strengthens criterion validity (clusters differ on variables *not used* to form them).

3.8. Complete Cases

The MEANS Procedure

```
20 proc sql noprint;
21   select count(*) into :n_after trimmed from work.CHA_CC6;
22   quit;
23
24   %let n_removed = %eval(&n_before - &n_after);
25   %let pct_kept = %sysvalf(&n_after/&n_before);
26   %let pct_rem = %sysvalf(&n_removed/&n_before);
27
28 data work.CC6_SUMMARY;
29   length Metric $50;
30   format Count comma12. Percent percent8.2;
31   Metric="Total rows (dedup)"; Count=&n_before.; Percent=.; output;
32   Metric="Complete-case rows (&n drivers)"; Count=&n_after.; Percent=&pct_kept.; output;
33   Metric="Removed due to missing (&n drivers)"; Count=&n_removed.; Percent=&pct_rem.; output;
34 run;
35
36 title "Complete-Case Filter Summary";
37 proc print data=work.CC6_SUMMARY noobs; var Metric Count Percent; run;
38 title;
```

Variable	N	N Miss
dist_sea_z	2958	0
dist_waterway_z	2958	0
grond_z_z	2958	0
area_log_z	2958	0
price_log_z	2958	0
height_z	2958	0

Figure 25. Code snippet for complete cases and its results (Assign2/2_DataPreparation/7_CompleteCases.sas)

Each of the six drivers had <20% missing individually, so removing rows with any missing among them is a simple, transparent way to (i) ensure identical samples across Ward, k-means, and GMM; (ii) avoid injecting imputation assumptions that can distort distances; and (iii) keep interpretability tight for the assignment.

Kept rows (means ± sd)

The MEANS Procedure

Variable	Mean	Std Dev
dist_sea_z	0.0052647	0.9975646
dist_waterway_z	-0.0137741	0.9206658
area_log_z	0.0519317	0.9217295

Dropped rows (means ± sd)

The MEANS Procedure

Variable	Mean	Std Dev
dist_sea_z	-0.0899164	0.9709346
dist_waterway_z	0.0934471	1.1639396
area_log_z	0.0837330	1.0694494

Pearson Correlation Coefficients, N = 2958 Prob > |r| under H0: Rho=0

	dist_sea_z	dist_waterway_z	grond_z_z	area_log_z	price_log_z	height_z
dist_sea_z	1.00000	-0.37817 <.0001	-0.13883 <.0001	0.14428 <.0001	0.24719 <.0001	-0.10638 <.0001
dist_waterway_z	-0.37817 <.0001	1.00000	0.00214 0.9073	-0.04002 0.0295	-0.17565 <.0001	-0.02199 0.2318
grond_z_z	-0.13883 <.0001	0.00214 0.9073	1.00000	-0.15942 <.0001	-0.12590 <.0001	-0.00460 0.8027
area_log_z	0.14428 <.0001	-0.04002 0.0295	-0.15942 <.0001	1.00000	0.14505 <.0001	0.19588 <.0001
price_log_z	0.24719 <.0001	-0.17565 <.0001	-0.12590 <.0001	0.14505 <.0001	1.00000	0.07490 <.0001
height_z	-0.10638 <.0001	-0.02199 0.2318	-0.00460 0.8027	0.19588 <.0001	0.07490 <.0001	1.00000

Figure 26. Result of checking shape on Complete Cases (Assign2/2_DataPreparation/8_CCCheck.sas)

After filtering to records with no missing values on the six drivers, all six drivers show 0 missing post-filter. The kept vs dropped comparison indicates only small mean shifts in coastal geometry (e.g., dist_sea_z kept ≈ 0.005 vs dropped ≈ -0.090;

dist_waterway_z kept ≈ -0.014 vs dropped ≈ 0.093 ; area_log_z kept ≈ 0.052 vs dropped ≈ 0.084), suggesting the complete-case filter did not materially bias the spatial/size signal. Outliers are rare ($|z| > 4$ counts: sea=7, waterway=12, elevation=44, area=5, price=8, height=3) and acceptable to keep. Pairwise correlations remain modest (largest magnitude ≈ -0.378 between sea and waterway), and regression VIFs for every driver sit ~ 1.0 – 1.26 , confirming no multicollinearity concerns.

4. Data Analyses

4.1. Univariate Exploration of Clustering Drivers - Histograms

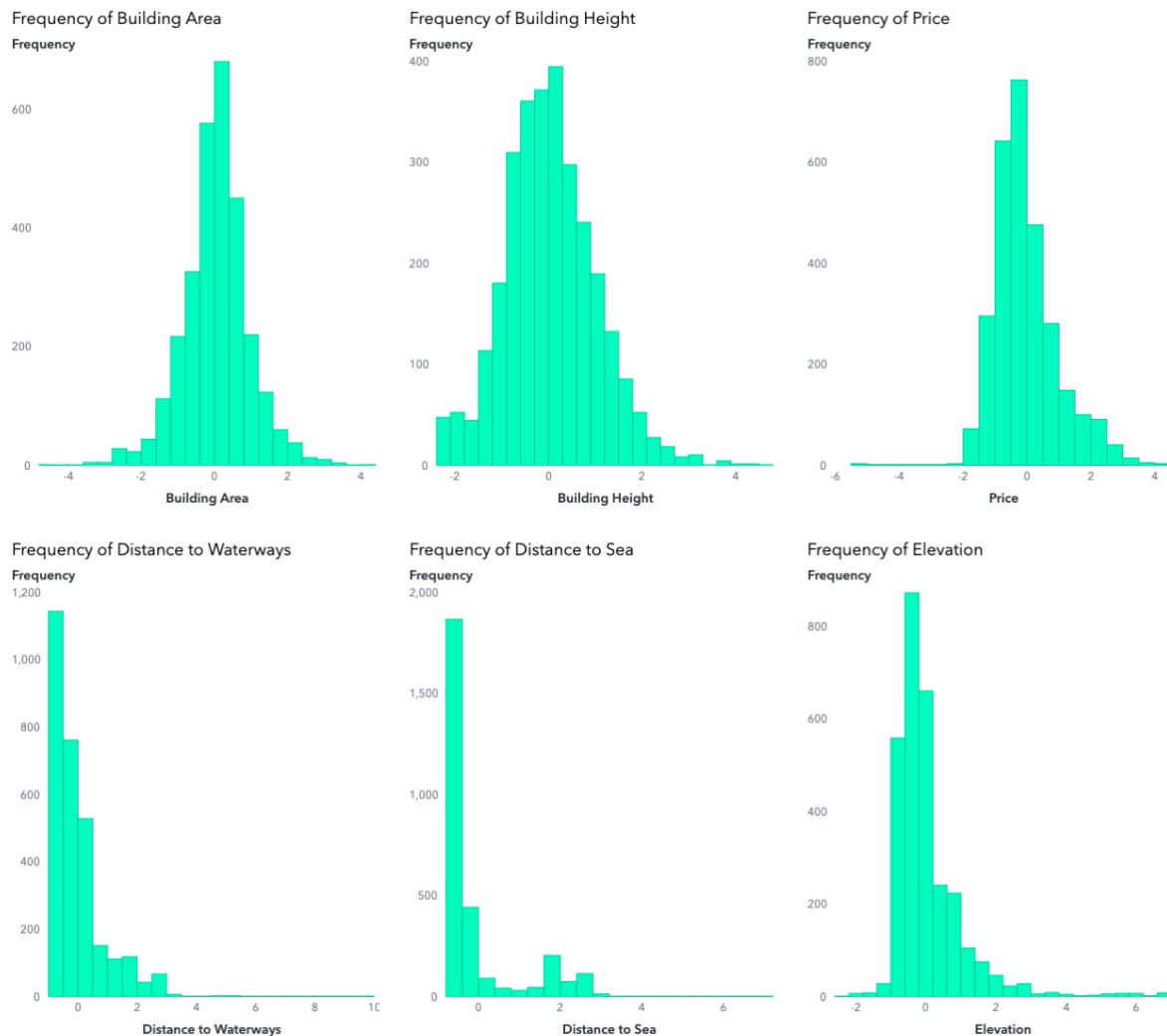


Figure 27. Histograms of Clustering Drivers

Across the six standardized drivers, Building Area, Building Height, and Price (area_log_z, height_z, price_log_z) are approximately normal and centered near zero. This is evidence that the log and z-score transformations worked and that these measures contribute comparably to Euclidean distances. In contrast, the spatial proximity variables – Distance to Sea, Distance to Waterways, and Elevation (dist_sea_z, dist_waterway_z, grond_z_z) are right-skewed, indicating most properties are close to coasts/canals and at low ground, with a smaller tail far inland or higher elevation. Together, these shapes justify the variable set: they capture two complementary dimensions which are location advantage (proximity/elevation) and

asset scale & pricing (area/height/price), which is expected to drive the heterogeneity revealed by clustering.

4.2. Univariate Exploration of Clustering Drivers – Box Plots

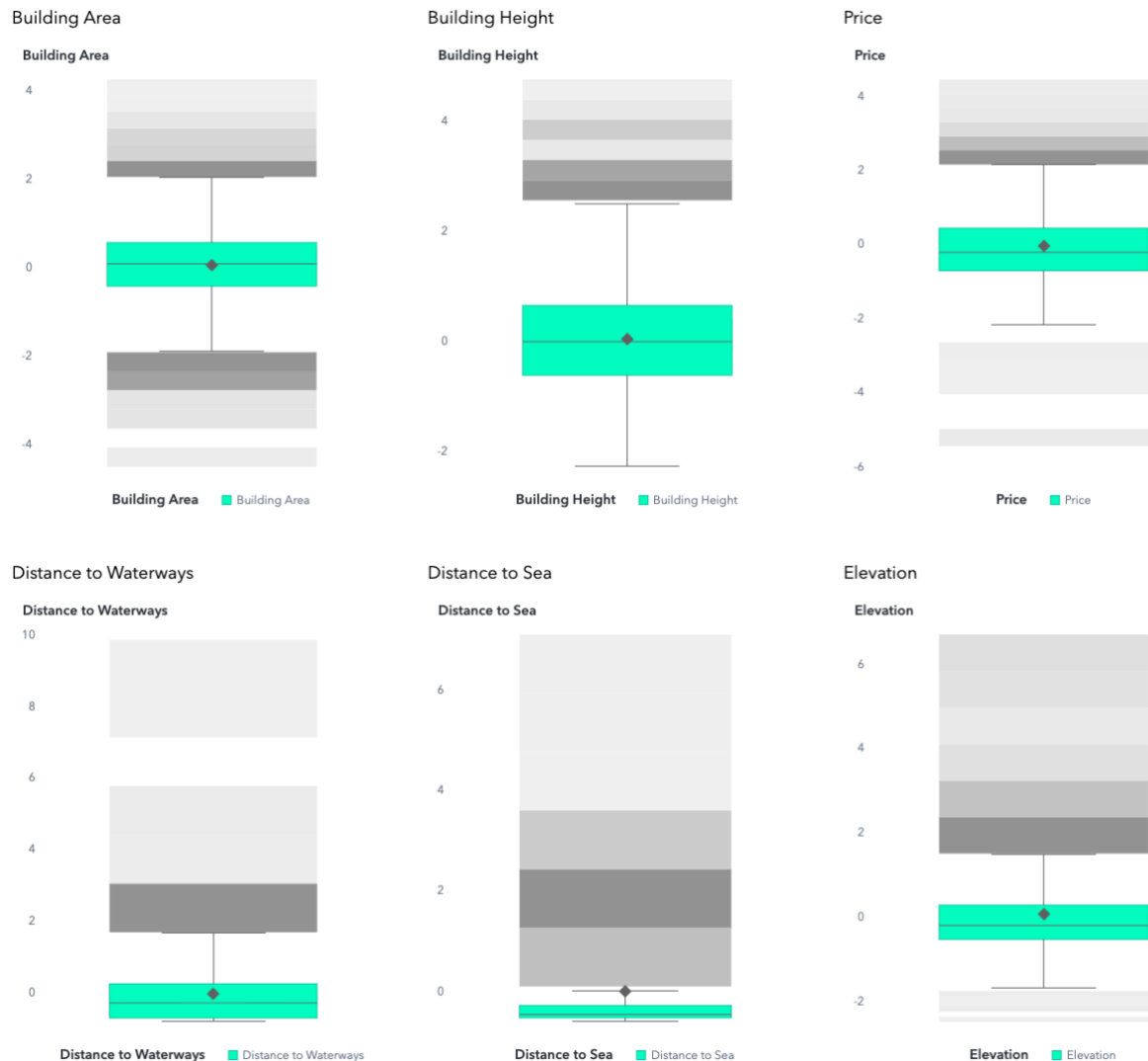


Figure 28. Box plots of clustering drivers

Overall, the boxplots confirm significant diversity in geographical and physical attributes, justifying the use of clustering to uncover natural groupings in the accommodation landscape. The right-skewed spatial variables (*distance to sea/waterways* and *elevation*) likely drive separation between coastal and inland clusters, while architectural and pricing variables (*area*, *height*, *price*) refine differentiation within those regions (e.g., premium vs. compact properties).

4.3. Tying in Univariate Analyses

```

1 ods select Moments Quantiles;
2 proc univariate data=casuser.CHA_CC6 vardef=df;
3   var area_log_z height_z price_log_z dist_waterway_z dist_sea_z grond_z_z;
4 run;
5
6 proc sql;
7   select
8     mean(dist_sea_z<=-0.5)*100 as pct_sea_le_0p5 format=6.2,
9     mean(dist_waterway_z<=-0.5)*100 as pct_wtr_le_0p5 format=6.2,
10    mean(grond_z_z<=0)*100 as pct_low_ground format=6.2
11  from casuser.CHA_CC6;
12 quit;

```

Figure 29. Code snippet for shape, symmetry and salience of univariate analyses

The UNIVARIATE Procedure Variable: area_log_z				The UNIVARIATE Procedure Variable: dist_waterway_z			
Moments				Moments			
N	2958	Sum Weights	2958	N	2958	Sum Weights	2958
Mean	0.05193174	Sum Observations	153.614089	Mean	-0.0137741	Sum Observations	-40.743795
Std Deviation	0.92172951	Variance	0.84958529	Std Deviation	0.92066583	Variance	0.84762557
Skewness	-0.0999315	Kurtosis	1.99067374	Skewness	2.36792096	Kurtosis	10.1148301
Uncorrected SS	2520.20115	Corrected SS	2512.22371	Uncorrected SS	2506.99001	Corrected SS	2506.4288
Coeff Variation	1774.88661	Std Error Mean	0.01694745	Coeff Variation	-6684.035	Std Error Mean	0.01692789

The UNIVARIATE Procedure Variable: price_log_z				The UNIVARIATE Procedure Variable: dist_sea_z			
Moments				Moments			
N	2958	Sum Weights	2958	N	2958	Sum Weights	2958
Mean	-0.0366087	Sum Observations	-108.28848	Mean	0.0052647	Sum Observations	15.5729687
Std Deviation	1.01775418	Variance	1.03582357	Std Deviation	0.99756463	Variance	0.99513519
Skewness	0.7643935	Kurtosis	2.04544854	Skewness	2.07296488	Kurtosis	4.13733261
Uncorrected SS	3066.89458	Corrected SS	3062.93028	Uncorrected SS	2942.69676	Corrected SS	2942.61477
Coeff Variation	-2780.0897	Std Error Mean	0.01871302	Coeff Variation	18948.1931	Std Error Mean	0.0183418

The UNIVARIATE Procedure Variable: height_z				The UNIVARIATE Procedure Variable: grond_z_z			
Moments				Moments			
N	2958	Sum Weights	2958	N	2958	Sum Weights	2958
Mean	0.04276355	Sum Observations	126.494595	Mean	0.07089686	Sum Observations	209.712922
Std Deviation	0.99233402	Variance	0.9847268	Std Deviation	1.06376317	Variance	1.13159208
Skewness	0.36759931	Kurtosis	0.73403917	Skewness	2.8331467	Kurtosis	11.6106058
Uncorrected SS	2917.24651	Corrected SS	2911.83715	Uncorrected SS	3360.98576	Corrected SS	3346.11778
Coeff Variation	2320.51339	Std Error Mean	0.01824563	Coeff Variation	1500.43756	Std Error Mean	0.01955896

Figure 30. Result of findings to tie-in with Univariate Analyses

The shape and symmetry statistics corroborate the patterns seen in the histograms and boxplots. The structural variables: building area, height, and price are approximately symmetric and well-normalized after the log and z-score transformations. Skewness values near zero (-0.10 , $+0.37$, $+0.76$ respectively) and moderate kurtosis (≈ 2.0) confirm these variables are roughly bell-shaped and evenly centered around the mean. This ensures that size, height, and price contribute proportionately to Euclidean distance in clustering, without distortion from extreme outliers.

In contrast, the spatial proximity variables: distance to waterways, distance to sea, and elevation exhibit strong positive skewness (≈ 2.1 – 2.8) and high kurtosis (>10). These long-right tails confirm that most accommodations are close to the coast or canals and at low elevation, while a smaller fraction sit far inland or higher above sea level. The salience analysis quantifies this: about 33% of listings are within 0.5 SD closer to the

sea than average, 39% are similarly close to waterways, and 64% lie at or below mean elevation. Together, these results reinforce the visual evidence that the dataset is dominated by low-elevation, near-water properties, yet retains meaningful variation in both location and structure where variation that the upcoming clustering analyses will exploit to reveal distinct accommodation segments.

4.4. Bivariate Analysis

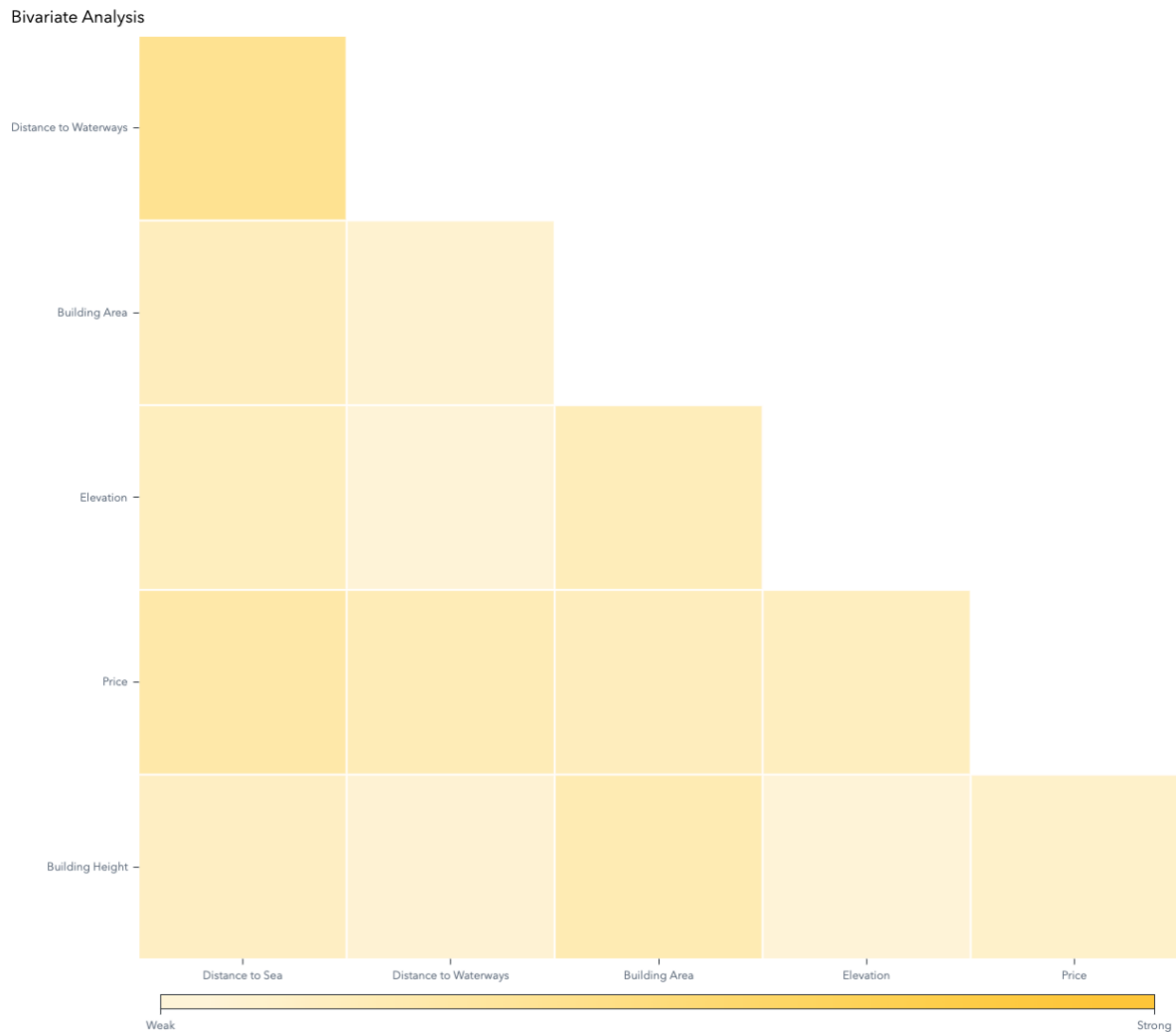


Figure 31. Correlation Matrix of Clustering Drivers

X Axis	Y Axis	Correlation
Distance to Sea	Distance to Waterways	-0.3782
Distance to Waterways	Price	-0.1756
Building Area	Elevation	-0.1594
Distance to Sea	Elevation	-0.1388
Elevation	Price	-0.1259
Distance to Sea	Building Height	-0.1064
Distance to Waterways	Building Area	-0.0400
Distance to Waterways	Building Height	-0.0220
Elevation	Building Height	-0.0046
Distance to Waterways	Elevation	0.0021
Price	Building Height	0.0749
Distance to Sea	Building Area	0.1443
Building Area	Price	0.1451
Building Area	Building Height	0.1959
Distance to Sea	Price	0.2472

Figure 32. Correlation values between clustering drivers

In the bivariate matrix of the six standardized drivers (N=2,958; Pearson on z-scores), the strongest association is between Distance to Sea and Distance to Waterways ($r = -0.3782$), a moderate, negative relationship indicating that locations nearer the sea tend to be farther from waterways, and vice-versa. All other pairs are weak ($|r| \leq 0.25$), implying little risk of redundancy for clustering: Distance to Sea vs Price $r \approx +0.247$ (slightly higher prices farther from the sea), Distance to Waterways vs Price $r \approx -0.176$ (slightly higher prices nearer waterways), Area vs Height $r \approx +0.196$ (larger sites are modestly taller), Elevation vs Area $r \approx -0.159$, and Elevation vs Distance to Sea $r \approx -0.139$. Remaining pairs are near zero (e.g., Elevation vs Height $r \approx -0.005$). Overall, the matrix shows no high collinearity, supporting the use of all six drivers in clustering without additional variable pruning.

4.5. Hierarchical Clustering: Cluster Observation

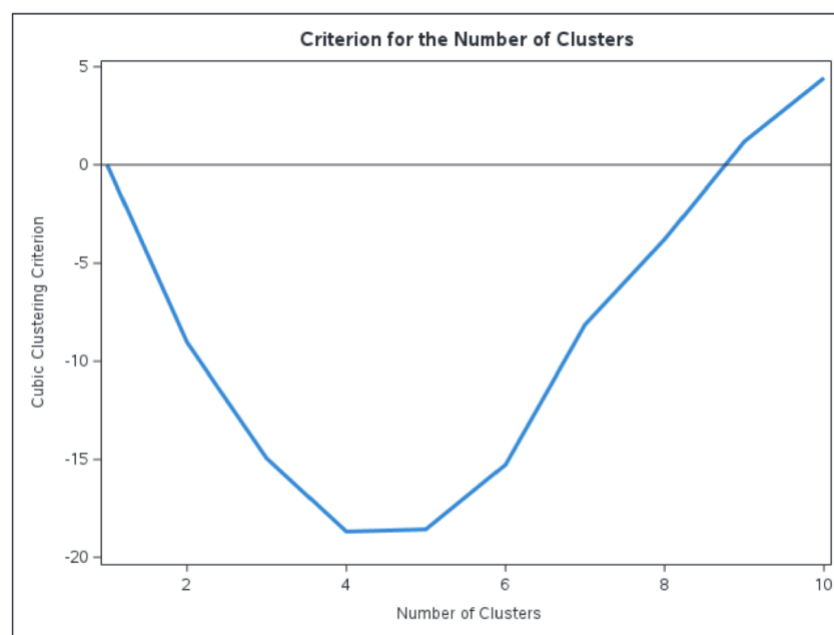


Figure 33. Cubic Clustering Criterion with the 6 Clustering Variables

Cluster History							
Number of Clusters	Clusters Joined		Freq	Semipartial R-Square	R-Square	Approximate Expected R-Square	Cubic Clustering Criterion
10	CL32	CL26	322	0.0181	.579	.569	4.43
9	CL14	CL16	545	0.0231	.556	.554	1.19
8	CL19	CL10	790	0.0299	.526	.535	-3.8
7	CL12	CL21	441	0.0319	.495	.514	-8.1
6	CL27	CL11	874	0.0467	.448	.488	-15
5	CL9	CL51	594	0.0503	.398	.451	-19
4	CL8	CL6	1664	0.0550	.343	.402	-19
3	CL5	CL13	853	0.0719	.271	.320	-15
2	CL4	CL3	2517	0.0909	.180	.209	-9.0
1	CL2	CL7	2958	0.1799	.000	.000	0.00

Figure 34. Results of Cluster Observations(Assign2/3_ExploreVisualize/1_HierarchicalClustering.ctl)

The CCC was evaluated on the observation-level Ward solution to choose a defensible cut. CCC is positive at 10 and 9 clusters(4.43 and 1.19), but turns negative from 8 downward, indicating weak global tendency toward a single “natural” k and motivating a pragmatic cut guided by business interpretability. For context, the root-mean-square

inter-observation distance is 3.4189 in the standardized six-driver space. Looking at Figure 33, it was concluded to adopt $k=5$ to preserve salient structure while avoiding over-fragmentation.

4.6. Hierarchical Clustering: Dendrogram

```

4      /* 1) Bin coordinates into quantiles */
5  ⊖  proc rank data=casuser.CHA_CC6 groups=&NQ out=work._grid;
6      var latitude longitude;          /* change names if needed */
7      ranks lat_q lon_q;
8      run;
9
10 ⊖  data work._grid;
11     set work._grid;
12     length region_id $8;
13     region_id = cats('G', put(lat_q, z2.), ' ', put(lon_q, z2.));
14     run;
15
16     /* 2) Centroids by grid region */
17  ⊖  proc means data=work._grid nway noprint;
18     class region_id;
19     var dist_sea_z dist_waterway_z grond_z_z area_log_z price_log_z height_z;
20     output out=work._grid_centroids(drop=_type_) mean=/ autoname;
21     run;
22
23     /* 3) Ward on grid centroids (weighted), then plot dendrogram */
24     ods graphics on / reset=all width=1200px height=800px imagename="cc6_grid_dendro";
25  ⊖  proc cluster data=work._grid_centroids method=ward ccc pseudo print=15
26     outtree=work._grid_tree
27     plots(only)=(ccc dendrogram(height=rsq));
28     var dist_sea_z_mean dist_waterway_z_mean grond_z_z_mean
29     area_log_z_mean price_log_z_mean height_z_mean;
30     id region_id;
31     freq _FREQ_;
32     run;

```

Figure 35. Code chunk to bin coordinates into quantiles for dendrogram
(Assign2/3_ExploreVisualize/2A_Dendrogram.sas)

To make the hierarchical tree legible for 2958 observations, each property was assigned to a 10x10 latitude-longitude quantile grid and used the grid ID as the dendrogram leaf label. Each dendrogram leaf represents a cell's median attribute values. This binning greatly reduced the number of leaf nodes and gave each leaf a **spatial meaning** (each has a defined geographic bounding box and a count of listings). Such an approach preserves clustering fidelity while yielding a more interpretable tree.

Obs	region_id	_FREQ_	n_props	lat_mean	lon_mean	lat_min	lon_min	lat_max	lon_max
1	G00_08	11	11	44.0023	12.6524	43.9921	12.6459	44.0055	12.6544
2	G00_09	284	284	43.9885	12.6825	43.9637	12.6547	44.0054	12.7512
3	G01_07	93	93	44.0306	12.6207	44.0160	12.6078	44.0402	12.6387
4	G01_08	193	193	44.0111	12.6472	44.0057	12.6389	44.0179	12.6542
5	G01_09	11	11	44.0063	12.6555	44.0055	12.6545	44.0073	12.6572
6	G02_05	7	7	44.0903	12.5321	44.0863	12.5314	44.0914	12.5328
7	G02_06	252	252	44.0712	12.5675	44.0517	12.5329	44.0916	12.5972
8	G02_07	36	36	44.0443	12.6045	44.0402	12.5977	44.0522	12.6113
9	G03_04	10	10	44.1446	12.4668	44.1371	12.4614	44.1460	12.4680
10	G03_05	284	284	44.1197	12.4979	44.0918	12.4680	44.1460	12.5328
11	G03_06	2	2	44.0917	12.5333	44.0916	12.5333	44.0917	12.5333
12	G04_03	33	33	44.2065	12.3949	44.2010	12.3898	44.2085	12.4007
13	G04_04	261	261	44.1735	12.4339	44.1463	12.4010	44.2049	12.4680
14	G04_05	2	2	44.1462	12.4684	44.1460	12.4682	44.1465	12.4687
15	G05_00	8	8	44.4112	12.3104	44.4052	12.3086	44.4138	12.3125
16	G05_01	2	2	44.3266	12.3369	44.3266	12.3369	44.3266	12.3370
17	G05_02	166	166	44.2885	12.3479	44.2609	12.3391	44.3157	12.3525
18	G05_03	120	120	44.2492	12.3651	44.2089	12.3527	44.2760	12.3952
19	G06_00	256	256	44.9686	12.2748	44.4188	12.2277	45.3156	12.3140
20	G06_01	3	3	45.3435	12.3203	45.3329	12.3193	45.3488	12.3225

Figure 36. Result of the binning of latitude and longitude.

Each label has a centroid and bounding box plus a count of listings which makes the dendrogram leaf traceable to real geographic cell and shows how representative it is.

Proximity Tag	Region IDs (comma-separated)
Away from Sea & Near Waterways	G07_00, G07_01, G07_02, G07_03, G08_00, G08_01, G08_02, G08_03, G08_04, G09_00, G09_01, G09_04
Near Sea & Away from Waterways	G01_07, G02_05, G02_06, G02_07, G03_05, G03_06, G05_00, G05_03, G09_07, G09_08
Near Sea & Near Waterways	G00_08, G00_09, G01_08, G01_09, G03_04, G04_03, G04_04, G04_05, G05_01, G05_02, G06_00, G06_01, G06_02, G06_03, G08_05, G08_06, G09_06

Figure 37. Label and Proximity Tag of Dendrogram

The binned leaves take on each region's median-z distance to the sea and to waterways(negative = closer than average) and assigning one of three observed environmental regimes:

1. Near Sea & Near Waterways,
2. Near Sea & Away from Waterways, and
3. Away from Sea & Near Waterways

The grouped label table shows 17 regions in category, as percentages: 43.6% in category 1, 25.6% in category 2 and 30.8% in category 3. This distribution confirms that the study area is dominated by (1)coastal-waterway adjacency with substantial representation of (2)open-coast and (3)inland-waterway.

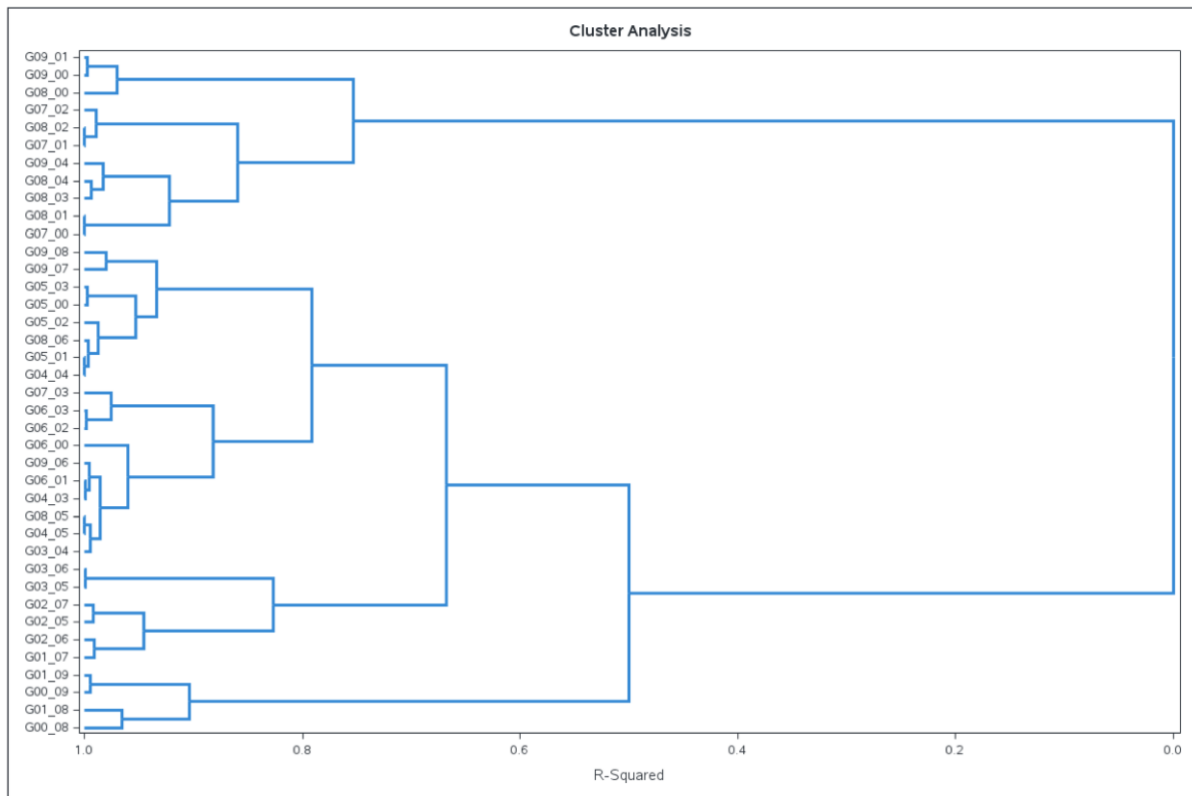


Figure 38. Dendrogram of the binned latitude and longitude vs the 6 cluster variables

Eigenvalues of the Covariance Matrix				
	Eigenvalue	Difference	Proportion	Cumulative
1	1.62486118	0.46886896	0.2780	0.2780
2	1.15599222	0.09941666	0.1978	0.4758
3	1.05657556	0.25792086	0.1808	0.6566
4	0.79865470	0.10244993	0.1367	0.7932
5	0.69620478	0.18400471	0.1191	0.9124
6	0.51220006		0.0876	1.0000

Root-Mean-Square Total-Sample Standard Deviation 0.986956

Root-Mean-Square Distance Between Observations 3.418915

Figure 39. Result from the Cluster Observation Task

The dendrogram provides an interpretable story: the five-cluster cut partitions the grid leaves so that large blocks of Near Sea & Near Waterways regions sit together, reflecting shared exposure characteristics along the lagoonal/coastal interface, while Near Sea & Away from Waterways and Away from Sea & Near Waterways form their own coherent branches rather than being scattered, preserving meaningful environmental contrasts at $k=5$.

It is insightful to note how much variance in the data is captured by the first few dimensions of this hierarchical solution. A principal components analysis of the same Ward clustering run shows that the first two principal components explain about 47.6% of the total variance, and the first three explain 65.7%. This implies that much of the cluster separation can be visualized in two or three dimensions, consistent with the clear geographic story observed. In summary, the Ward dendrogram (with spatial binning) suggests a five-cluster structure where each cluster corresponds to a distinct

coastal exposure archetype. This sets the stage for the next step of confirming and profiling these clusters with non-hierarchical methods.

4.7. K-Means Clustering

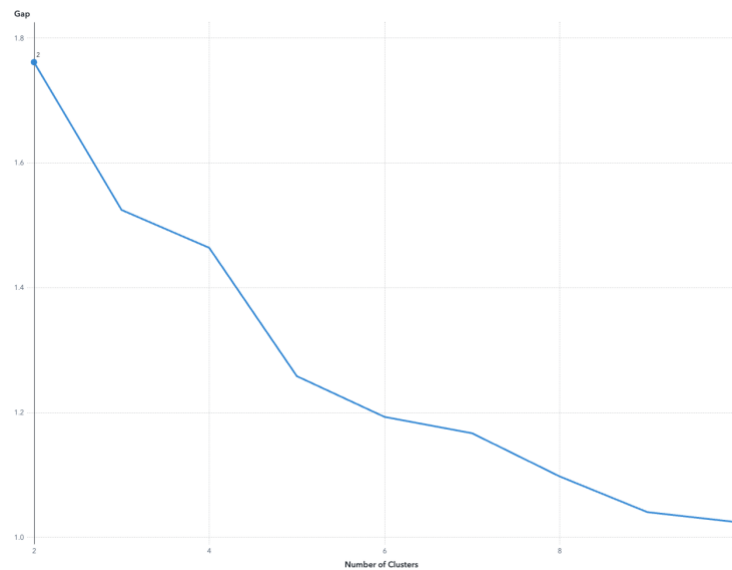


Figure 40. Aligned Box Criterion for K-Means Clustering

The aligned box criterion (a gap-style heuristic) peaks at $k=2$, indicating a strong global separation; however, per SAS guidance to prioritize interpretability, it is important to examine Ward's hierarchy with CCC, pseudo-F, pseudo- t^2 , and the dendrogram. Those diagnostics show a prominent merge change around the $5 \rightarrow 4$ cut, and the $k=5$ k-means delivers balanced and well-separated segments with clear business-meaningful profiles.

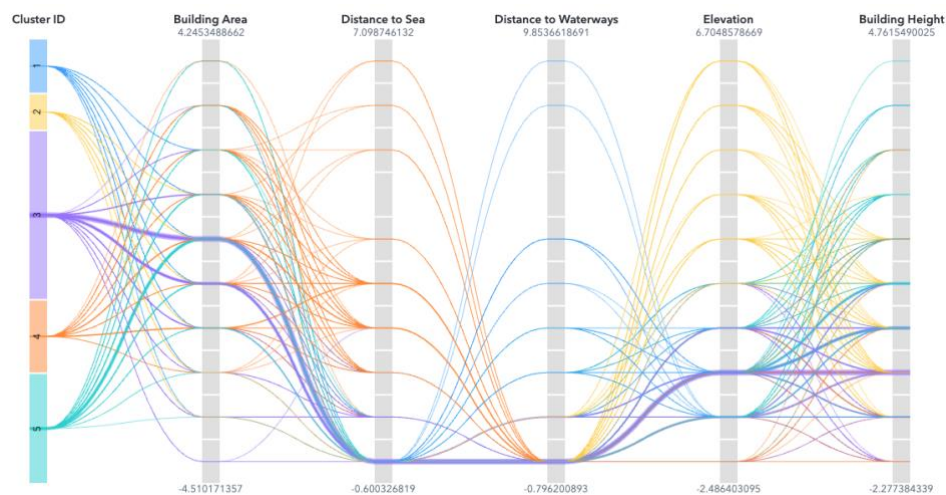
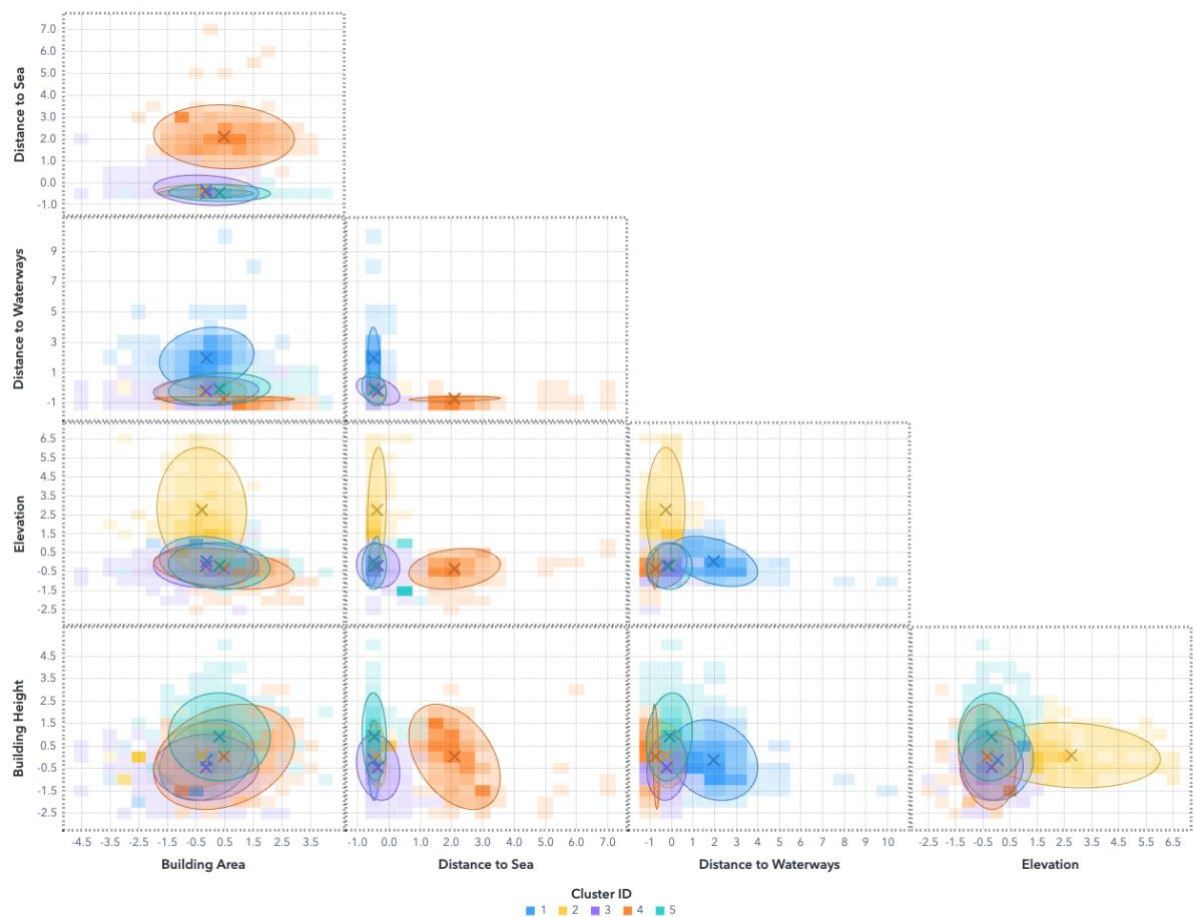


Figure 41. Parallel coordinates of K-means clustering.



Cluster ID	Observations	RMS of STD	Within cluster SS	Min centroid-to-observation	Max centroid-to-observation	Nearest Cluster	Centroid Distance	Average Distance
1	364	1.7694203554	1139.6288154	0.349598238	8.0732478456	3	2.2147165126	1.6066144266
2	246	2.0376795518	1021.4259371	0.3985016067	4.8736544496	3	2.9792412844	1.8453501158
3	1,125	1.5138581568	2578.2373337	0.2553085333	5.2298344303	5	1.7753284402	1.3762508512
4	488	2.0199508077	1991.1382177	0.3981269087	5.3600280293	5	2.8004293518	1.8661772146
5	735	1.8202610528	2435.3124708	0.2533097752	4.6633062103	3	1.7753284402	1.6680942038

Figure 42. K-means clustering of the 6 clustering variables

All variables are z-scores; hence, negative values indicate closer/smaller/shorter/cheaper than the sample mean, positive values are opposite.

Cluster 1: Near-sea, away from rivers, budget low-rise (N=364).

Cluster 1 is driven by Sea = -0.48 and Waterways = +1.96 (closest to sea, far from waterways), plus Price = -0.45 and Height = -0.14 (value-priced, low-rise). Area is slightly below average (-0.14); Elevation near zero (+0.05). Compactness is reasonable (average distance to centroid 1.61) with a few far points (max distance ~8.07).

Cluster 2: High-ground coastal, small, value (N=246).

Cluster 2 follows the standout Elevation = +2.75 alongside Sea = -0.39 and Waterways = -0.28 (coastal despite high ground). Structure and price are smaller/value (Area = -0.31, Price = -0.38, Height ≈ +0.09). This is one of the looser clusters (avg distance 1.85).

Cluster 3: Waterfront budget low-rise (mass segment) (N=1,125).

Cluster 3 follows Sea = -0.35 and Waterways = -0.22 (close to both) and the lowest Price = -0.50 with Height = -0.45 (budget, low-rise). Area slightly below average

(-0.16); Elevation slightly below (-0.17). It is the tightest cluster (avg distance 1.38), and its nearest centroid is Cluster 5 (distance ≈ 1.78), consistent with the scatter-matrix ellipses.

Cluster 4: Canal-front inland premium, larger footprint (N=488).

Cluster 4 for Sea = +2.10 (farthest from sea) and Waterways = -0.74 (very close to rivers/canals), combined with Area = +0.47 and Price = +0.64 (larger, premium). Height is near zero (+0.01), Elevation below average (-0.33). This group is among the looser ones (avg distance 1.87); its closest neighbor is Cluster 5 (centroid separation ≈ 2.80).

Cluster 5: Beach-near premium high-rise (N=735).

Cluster 5 for Sea = -0.47 (near sea) and the tallest Height = +0.90 with Price = +0.54 and Area = +0.31 (premium coastal towers). Waterways proximity is mild (-0.11), Elevation slightly below (-0.18). Compactness is solid (avg distance 1.67); nearest centroid is Cluster 3 (~ 1.78), indicating a premium vs budget continuum within near-water stock.

The five segments are balanced and cover the full supply spectrum: C1 to C5 percentages are 12.3%, 8.3%, 38.0%, 16.5%, and 24.9%, respectively. Compactness is acceptable across groups: the average distance from observations to their cluster centroid ranges from 1.376 (C3, tightest) to 1.866 (C4, loosest), with C1=1.607, C2=1.845, and C5=1.669. The smallest centroid-to-centroid separation occurs between C3 and C5 (1.775), indicating a smooth budget to premium coastal continuum rather than a hard break; the largest separation is C2 vs C3 (2.979), showing that the high-ground coastal value niche is most distinct from the waterfront budget mass segment.

4.8. Gaussian Mixture Model

Input Variable Statistics				
Variable	Mean	Standard Deviation	Min	Max
grond_z_z	0.070897	1.063763	-2.486403	6.704858
dist_sea_z	0.005265	0.997565	-0.600327	7.098746
dist_waterway_z	-0.013774	0.920666	-0.796201	9.853662
area_log_z	0.051932	0.921730	-4.510171	4.245349
height_z	0.042764	0.992334	-2.277384	4.761549
price_log_z	-0.036609	1.017754	-5.449908	4.458907

Cluster Information													
Cluster ID	Cluster Weight	grond_z_z Mean	dist_sea_z Mean	dist_waterway_z Mean	area_log_z Mean	height_z Mean	price_log_z Mean	grond_z_z Variance	dist_sea_z Variance	dist_waterway_z Variance	area_log_z Variance	height_z Variance	price_log_z Variance
1	0.0646012739	-0.17343768	-0.48778265	0.8591331249	-0.230522708	-0.072691357	0.4868287108	0.224206929	0.0125229279	1.428942902	2.2900109433	1.2511050484	2.8625375709
2	0.1516162091	0.1473070825	-0.471111149	1.2472093853	-0.034961215	-0.071910288	-0.579070945	0.3611260961	0.0090778897	0.9006875344	0.2497392712	0.6174227066	0.3126292463
3	0.1760112216	0.8912755898	-0.425595824	-0.250743696	-0.155320527	0.10999996	-0.521930569	0.9424375649	0.0086613041	0.1487360123	0.3829169011	0.7367975607	0.3438363198
4	0.0360077292	3.3294493094	-0.330512031	-0.264748387	-0.388412295	-0.049632336	0.2294543077	4.7247927495	0.054116649	0.1186248881	0.7443046366	0.4885540438	1.1923940736
5	0.0028861211	-0.621989843	-0.33907563	5.0096930411	0.2887666388	-0.855292675	-0.318127549	0.1753863073	0.0951506186	4.7409688794	0.3554754237	0.2150846576	0.2662803092
6	0.2012817161	-0.331148085	1.7942371947	-0.738704958	0.3459180957	-0.05551387	0.5246513868	0.2516318481	0.824577934	0.0060609039	1.2975563688	1.1798709383	0.8877282893
7	0.3180050024	-0.394307265	-0.504025711	-0.137234391	0.2382567403	0.3131880465	0.052769916	0.1055309494	0.0042556001	0.145611069	0.4606123787	1.010563083	1.0164657598
8	0.0495907266	-0.477104205	-0.092020802	-0.578703387	-0.654016393	-0.897349348	-0.364253341	0.0698299871	0.1236596785	0.0456662595	1.4845550137	0.3637140757	0.1447290332

Model Information	
Clustering Algorithm	Gaussian Mixture Model
Inference Method	Variational Bayesian
Random Number Seed	12345
Clusters Maximum	8
Dirichlet Process Mass	1
Covariance Matrix Type	Diagonal
VB Iterations	84
VB Threshold	0.001
VB Converge	Yes
Clusters Discovered	8

_PREDICTED_CLUSTER_	Frequency	Percent	Cumulative Frequency	Cumulative Percent
3	512	17.31	1046	35.36
4	90	3.04	1136	38.40
5	8	0.27	1144	38.67
6	584	19.74	1728	58.42
7	1081	36.54	2809	94.96
8	149	5.04	2958	100.00

Figure 43. Output of GMM (Assign2/3_ExploreVisualize/3_GMM.sas)

As a final step, a **Gaussian Mixture Model (GMM)** was fitted to the data to further validate the segmentation. The GMM is a model-based clustering approach that provides soft assignments (each listing has a probability of belonging to each cluster) and can automatically discover an appropriate number of clusters based on data complexity. GMM converged on an 8-cluster solution. Notably, these 8 clusters turned out to be largely **refinements or subdivisions** of the five k-means segments rather than entirely new groupings – a reassuring sign that the underlying structure is consistent.

- C7 (36.54%): nearer the sea ($\text{dist_sea_z} = -0.504$), slightly closer to waterways ($\text{dist_waterway_z} = -0.137$), larger area ($\text{area_log_z} = 0.238$), taller ($\text{height_z} = 0.313$), above-average price ($\text{price_log_z} = 0.313$). This is a sizeable, near-sea, mid-rise, modest-premium segment.
- C6 (19.74%): far from sea ($\text{dist_sea_z} = 1.794$), near waterways ($\text{dist_waterway_z} = -0.739$), larger ($\text{area_log_z} = 0.346$), but price about average/slightly below ($\text{price_log_z} = -0.056$). This looks like inland, canal-adjacent, larger-footprint stock without a strong price premium.
- C3 (17.31%): near sea (-0.426), slightly near waterways (-0.251), smaller area (-0.155), slightly taller (0.110), below-average price (-0.522): a coastal value segment.
- C2 (13.49%): near sea (-0.471), far from waterways (1.247), roughly neutral size and height (-0.035 , -0.072), below-average price (-0.579): another coastal value segment.
- Smaller clusters:
C1 (4.56%) near sea (-0.488), far from waterways (0.859), smaller (-0.231), price premium (0.487), a boutique near-sea premium niche. C4 (3.04%) very high elevation ($\text{grond_z_z} = 3.329$), near sea/waterway ($-0.331/-0.265$), small, slight premium (0.229). C8 (5.04%) near waterways (-0.579), small & short, discount (-0.364). C5 (0.27%) is tiny and atypical (e.g., $\text{dist_waterway_z} = 5.010$), suggesting an outlier pocket.

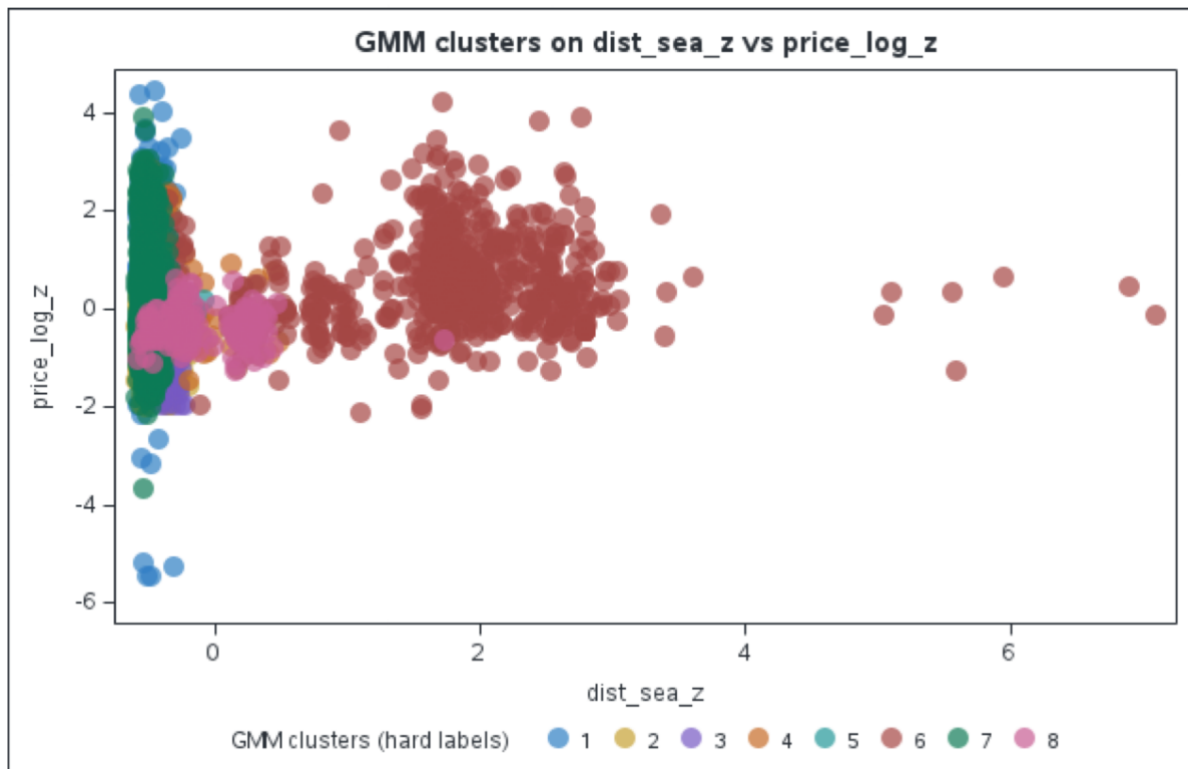


Figure 44. GMM Cluster Scatter Plot of Price (Log Transformed and Z-scored) vs Distance to Sea (Z-scored)

The cloud at near-sea ($x < 0$) contains both premium and discount listings, while the far-sea ($x > 0$) cloud is dominated by one cluster with prices around average. In other words, sea proximity alone does not guarantee a premium. The premium appears only when combined with certain *form-factor* attributes (area/height) and sometimes waterway proximity. This matches the cluster means: two near-sea clusters carry positive price deviations (C1: +0.487 SD; C7: +0.313 SD), while three other near-sea clusters are discount (C2: -0.579 SD; C3: -0.522 SD; C8: -0.364 SD).

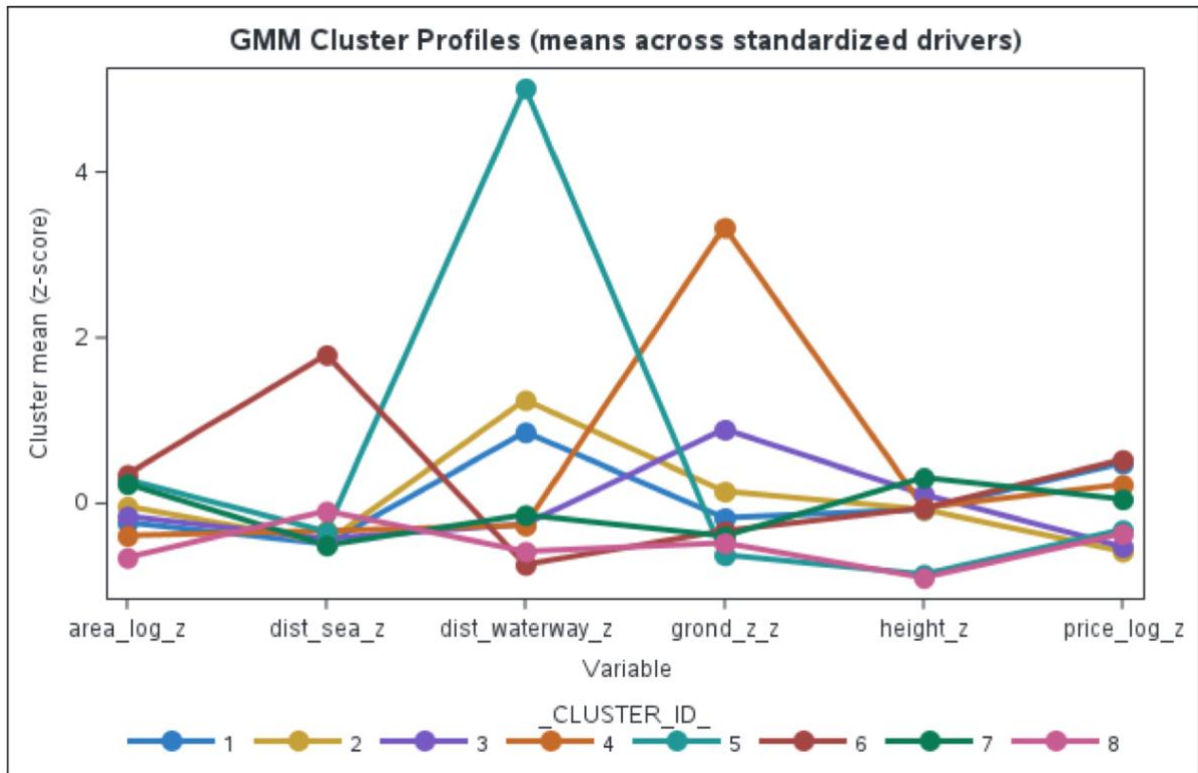


Figure 45. GMM Cluster Profiles

Analysis Variable : max_post					
N	Minimum	25th Pctl	Median	75th Pctl	Maximum
2958	0.2613256	0.7568861	0.9158026	0.9844300	1.0000000

Figure 46. GMM Cluster (Means across standardized drivers)

Posterior max-probability is high: median = 0.9158, IQR 0.7569–0.9844 (25th–75th), max = 1.0000 which indicates **confident assignments overall**. The radar chart (cluster means across standardized drivers) shows that geography organizes the market and form-factor stratifies price within it: clusters split first by sea and waterway proximity and by elevation, then diverge on area/height and price. Overall, the radar confirms that sea proximity alone is insufficient; near-sea + size/height (sometimes aided by waterway adjacency) is what consistently lifts price, while near-sea without those attributes settles into value tiers.

4.9. Price by Segment: What Each Cluster Commands

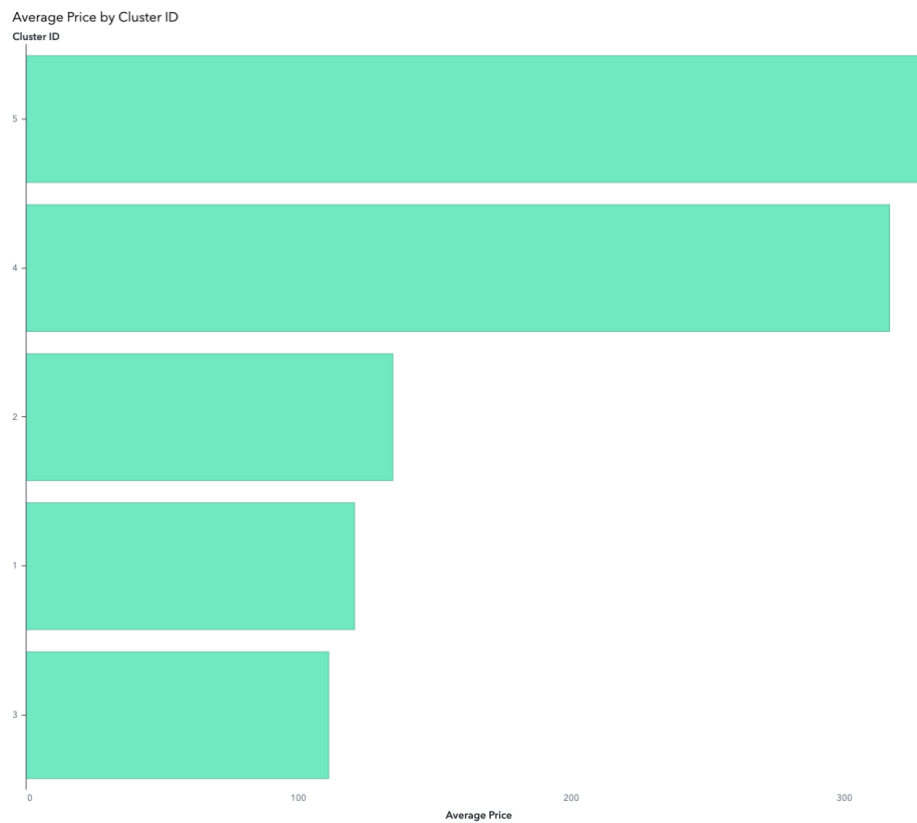


Figure 47. Cluster ID vs Average Prices

The bar chart is consistent with the clustering and working labels, showing compositional levers behind the monetisation. Cluster 5 (“beach-near premium high-rise”) and Cluster 4 (“canal-front inland premium, larger footprint”) monetise at the top end, while Clusters 1/3 (“budget low-rise” near sea/waterfront) anchor the lower tiers, and Cluster 2 (“high-ground coastal, small, value”) sits in between.

4.10. What Drives Price: Product Type & Access Features



Figure 48. Average Price by Access Features

Type of Room:

Mean price is highest for Holiday Rental Homes (≈ 284), then Camping (≈ 209) and B&B (≈ 206), with Hotels lower (≈ 185). Hostels sit at the bottom (≈ 65). Sample sizes are large for Hotels ($n=2,269$) and Holiday Rental Homes ($n=547$) but very small for Agriturismo ($n=4$) and Hostel ($n=6$), so treat those two with caution. A one-way ANOVA across Types is significant ($F \approx 9.91$, $p \approx 7.44 \times 10^{-11}$), indicating real differences in mean price between types.

Beach Access:

The mean is highest under “No” beach access (≈ 255), then “Other”/NA (≈ 210), “Yes (Private)” (≈ 172), and “Yes” (≈ 150). This tells us that direct beach access isn’t the only, or even primary price driver in this dataset; inland premium or canal-front locations (matching Cluster 4 label) likely command high prices without formal “beach access.” Differences are statistically significant ($F \approx 42.66$, $p \approx 5.46 \times 10^{-19}$).

Pool Access:

No pool and NA categories (≈ 209 – 212) average higher than shared pool (≈ 185), while private pool averages lowest (≈ 96) but with very small $n=19$, so that estimate is unstable. Differences are statistically significant but modest ($F \approx 3.22$, $p \approx 0.040$). Given the tiny private pool sample, it’s prudent not to over-interpret that category without more data.

5. Summary of Results

The hierarchical (Ward) analysis, k-means, and GMM collectively paint a coherent picture of the coastal hospitality market's structure. The Ward dendrogram provided the initial evidence for around five clusters and ensured those clusters respect the major geographic groupings (sea vs waterway exposure). The k-means implementation at $k=5$ then gave us stable, interpretable clusters with clear profiles and good separation. Finally, the GMM without forcing $k=5$ independently discovered segments that align with the same five-group structure, only adding nuance by separating tiny niche clusters. It is reassuring that a completely different technique (model-based vs. distance-based clustering) reached a congruent segmentation, which speaks to the robustness of these patterns (Jain, 2010). All three approaches indicate that geography is the primary organiser of listings into clusters, and building characteristics plus price define sub-segments within those geographical groups. The fact that GMM assignment probabilities are high further implies that these clusters are not artifacts of the algorithm, but real groupings in the data. In summary, using multiple methods has strengthened our confidence in the solution: the Ward method revealed the presence of five meaningful clusters, k-means concretely defined those clusters, and GMM confirmed them (while highlighting finer subclusters). We can thus proceed knowing our segmentation is valid, data-driven, and grounded in both domain knowledge and statistical evidence, a crucial consideration for subsequent risk analysis and strategic planning (IPCC, 2020; NHESS, 2021).

6. Managerial Recommendations

- **Wave and storm exposure:** Hotels on unsheltered reaches face higher runup and overtopping. Add long-term 99th-percentile significant wave height and/or extreme wave climate. Upon research, this might be readily available from the Copernicus Marine Mediterranean wave products (Copernicus Marine Service, 2021).
- **Land subsidence:** Relative SLR at the property is elevation change minus sea-level change. Northern Adriatic coasts (Veneto/Po Delta) are well known for subsidence; add vertical land motion from the European Ground Motion Service (EGMS) to quantify mm/yr sinking at or near each asset, then roll it into relative SLR rates. This has become straightforward since EGMS went operational. (Copernicus, n.d.)
- **Shoreline change & beach width:** Distance to water misses whether the shoreline is *advancing or retreating*. Add a shoreline change rate (m/yr) and beach width (peak season). Something like EU datasets derived from EUROSION (European Environment Agency, 2024) and the newer CoastTER database that integrates erosion indicators on shoreline baseline; they're European-scale and defensibly curated. Even a coarse sign (erosion vs accretion) improves separability.
- **Accessibility (time, not distance):** Replace or complement straight-line distances with travel time to the nearest beach access point, transit stop, town center, and nearest intercity rail station, computed on the road/walk network. These time-based variables are much more predictive for pricing and clustering than Euclidean distances.

- **Replace the duplicative “estuary distance” with a geomorphic class:** $\text{dist_estuary} \approx \text{dist_sea}$ for three-quarters of rows, switch to a coastal-setting categorical (open beach / lagoon or canalized back-beach / river mouth / hard-defended). It’s a single, interpretable splitter that captures different hazard pathways (overtopping vs backflow) and avoids double-counting distance.
- **Compact, cluster-friendly:** Clustering that reflects destination attractiveness without leaving the supply-side frame benefits from simple, reproducible proxies. Replace some straight-line distances with network travel times (e.g., to the nearest beach access point and intercity rail); add a small set of amenity densities (counts of restaurants, pharmacies, ATMs, marinas within 500–1,000 m from OSM). These measures are compact, auditable, and improve separation between “high-rise premium” and “waterfront budget” segments that happen to lie equally close to the sea.

7. References

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